

Session 4: Learning without labels, bias and governance, and how LLMs (large language models) work

Thursday 17 September 2026, 11:00 AM to 12:30 PM EDT (Eastern Daylight Time).
Videos V40 to V50.

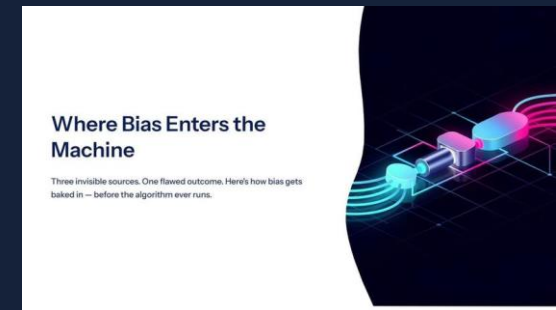
Student edition: the questions are here; the answers are discussed in the session.

Questions: Because of the interpretation delay, please send your questions in the Zoom chat, or better, after the session on the WhatsApp number you already have.

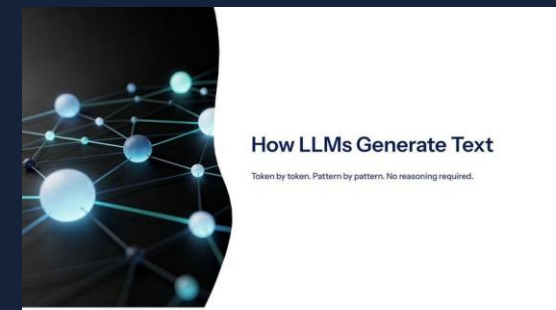
The interactive exercises are in the HTML (web page) edition used in class. In this file, each one is described instead, with its controls, what to try and its point.



V40 page 1: What Clustering Does



V44 page 1: Where Bias Enters the Machine



V48 page 1: How LLMs Generate Text

Block 1 of 11

What clustering does

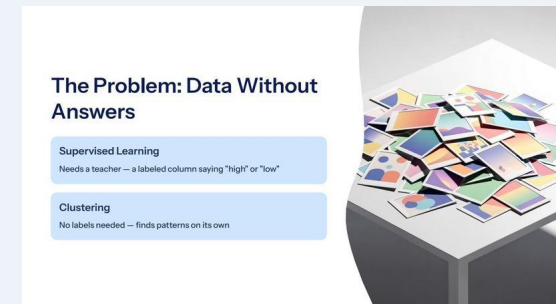
Video V40, What Clustering Does. 7 minutes.

Clustering puts similar records into groups when nobody has labelled them; people decide whether the groups are useful.

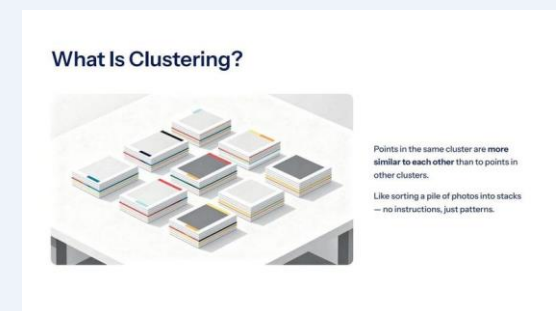
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 20 pages of V40.



V40 page 1: What Clustering Does



V40 page 2: The Problem: Data Without Answers



V40 page 3: What Is Clustering?

Clustering groups customers whose numbers are close, so every number has to be on the same scale first.

How far apart are A and B on the raw numbers?

Customer A is 35 and spends \$2,000 a year. Customer B is 36 and spends \$2,100. The gaps are 1 year and \$100, so spending makes up 99% of the difference. Customer C, aged 62 and spending \$2,050, looks closer to A than B does.

the word: **distance**

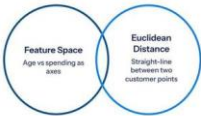
How far apart are they once each feature is scaled?

Each feature becomes points out of 100 over its own range: ages 20 to 70, spending \$0 to \$10,000. A to B is now 2 points for age and 1 point for spending, 3 in total. A to C is 54.5 points. B is the closest again.

the word: **feature scaling**

Scale first, then group. The algorithm cannot tell a big unit from a big difference.

How Similarity Is Measured



- Euclidean Distance**
- Each feature = one dimension in space
 - Distance = straight line between two points
 - Shorter distance → more similar customers

V40 page 5: How Similarity Is Measured

A Concrete Example

Customer A	Customer B
Age 35 · Spends \$2,000	Age 36 · Spends \$2,100

Without scaling: Distance ≈ 100 — the \$100 gap completely swamps the 1-year age gap.

V40 page 12: A Concrete Example

Scaling in Action

Before Scaling	After Scaling
Distance driven almost entirely by spending units Age is nearly invisible in the math	Both features pull equal weight Business priorities now reflected in the math

V40 page 13: Scaling in Action

Sorting the elements without an answer key, 1869

Step 1



About 63 elements, no order

In 1869 chemists knew about 63 elements. There was no agreed way to arrange them into families.

Step 2



One card per element

Dmitri Mendeleev, a professor in St Petersburg writing a chemistry textbook, wrote each element's weight and properties on a card.

Step 3



Similar cards side by side

He laid the cards out by weight and placed elements that behaved alike next to each other. No one handed him the groups.

Step 4



Gaps left open

Where no known element fitted, he left a blank space and described the weight and behaviour an element there should have.

Step 5



The groups prove useful

Gallium (1875), scandium (1879) and germanium (1886) were found and matched his descriptions closely. The table earned its place by what it let chemists do.

Sources: Chemistry World (Royal Society of Chemistry), "The father of the periodic table", which quotes Mendeleev's own account of the cards; Science History Institute, "An Element of Order".

Three things to keep

1

Clustering puts similar records into groups when no column holds the answer.

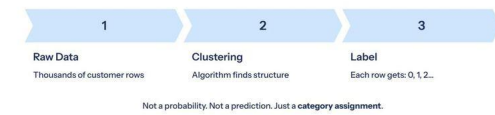
2

Scale every feature before grouping, or the feature with the biggest numbers decides who looks similar.

3

The result is a group number for each row; whether the groups are useful is a business call.

The Output: A Cluster Label



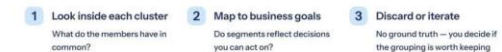
V40 page 6: The Output: A Cluster Label

Scaling Rule: Always First



V40 page 11: Scaling Rule: Always First

How to Evaluate Your Clusters



V40 page 17: How to Evaluate Your Clusters

A furniture retailer has 10,000 customer records with age and yearly spending, such as customer A: 35 years old, \$2,000 a year. No column says who is high value or at risk. The team runs clustering to find groups for marketing.

What does the clustering give back for customer A?

- A

A probability, such as 80%, that A is a high-value customer.
- C

A forecast of how much A will spend next year.

- B

A group number, such as group 2, with no score attached.
- D

A yes or no label copied from customers who were labelled before.

The Problem: Data Without Answers

- Supervised Learning

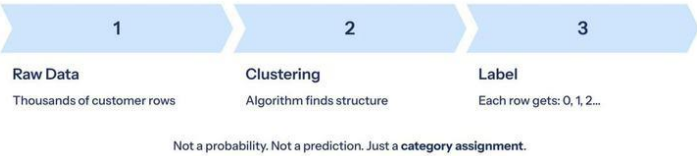
Needs a teacher — a labeled column saying "high" or "low"
- Clustering

No labels needed — finds patterns on its own



V40 page 2: The Problem: Data Without Answers

The Output: A Cluster Label



V40 page 6: The Output: A Cluster Label

Type the letter in the Zoom chat. Time to answer: 45 seconds.

The team compares three customers on the raw numbers, adding the age gap to the dollar gap. A is 35 and spends \$2,000. B is 36 and spends \$2,100. C is 62 and spends \$2,050.

Which customer looks most similar to A, and what does that tell the team?

A B, because B is only one year older than A.

B C, because \$50 is less than \$100; the dollar amounts are drowning out a 27-year age gap.

C B and C tie, because both spend about \$2,000.

D Neither, because age and money can never be compared.

The Distortion Problem

\$5,000 vs \$4,000
Looks **dramatically** far in distance calculations

Age 30 vs 31
Looks **incredibly** close — yet may matter more to your business

⚠ Large-number features dominate by accident — not because they're more important.

V40 page 8: The Distortion Problem

A Concrete Example

Customer A Age 35 · Spends \$2,000	Customer B Age 36 · Spends \$2,100
---------------------------------------	---------------------------------------

Without scaling: Distance ≈ 100 — the \$100 gap completely swamps the 1-year age gap.

V40 page 12: A Concrete Example

Type the letter in the Zoom chat. Time to answer: 60 seconds.

After scaling, the team runs clustering again and gets three groups. Looking inside, the groups mostly follow the region where customers live. Marketing wanted groups based on spending habits for a loyalty offer.

What should the team conclude?

A The algorithm made an error and should be replaced.

B The math has no mistakes, so the groups are right and should be used.

C The groups do not serve this goal; review the features and the goal, then run it again.

D Check the groups' accuracy against the true customer groups.

Useful vs. Useless Clusters



Useful

Clusters separate high-spend from low-spend — and that's what your business needs



Useless

Clusters separate by zip code — but geography wasn't your goal

Same algorithm. Same math. **Completely different business value.**

V40 page 15: Useful vs. Useless Clusters

The algorithm knows the math.
You know the business.

Examine each cluster. Ask: do these segments drive real decisions?

V40 page 16: The Algorithm Knows the Math; You Know the Business

Type the letter in the Zoom chat. Time to answer: 60 seconds.

Your turn: who looks most similar to whom?

Eight customers, each with an age and a yearly spending. The table ranks the other seven by total gap. Change the three settings on the left.

What you can change in the interactive

Customer to match

Options: A: 35 years, \$2,000; B: 36 years, \$2,100; C: 62 years, \$2,050; D: 50 years, \$5,250; E: 62 years, \$8,250; F: 27 years, \$5,900; G: 47 years, \$2,100; H: 22 years, \$9,450

The customer we are finding the closest match for.

Spending recorded in

Options: dollars; hundreds of dollars; thousands of dollars

The unit chosen when the spending column was stored. The customers do not change.

Scale both features to points out of 100

Tick box: off or on

Points out of 100 over each range: ages 20 to 70, spending \$0 to \$10,000.

Try, in this order

1. Scaling off: switch spending from dollars to thousands. A's closest match changes; no customer did.
2. Scaling on: change the unit again. Nothing moves.
3. Customer E: try dollars, then thousands, then scaling on. Three different closest matches.

The point: Without scaling, the answer depends on the unit someone picked. With scaling, it depends on the customers.

The eight customers are made-up values chosen to show the effect; A and B come from the video. Total gap adds the age gap and the spending gap. The straight-line distance used in the video picks the same closest match in every setting.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

1

With no answer column, clustering is how you explore customers. There is no accuracy to report; you report what the groups mean.

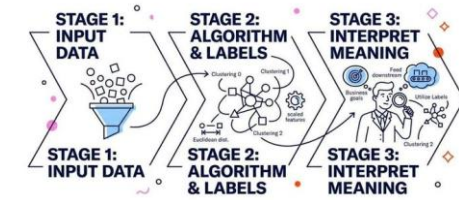
2

Scale before grouping. On raw dollars, C (62) was the closest match for A (35); on scaled points, B (36) is back in first place.

3

Judge each group by the decision it supports. Groups by region are useless when the offer is about spending.

Clustering: The Complete Picture



V40 page 18: Clustering: The Complete Picture



Success Is Measured by Usefulness

Not by accuracy on a test set. Not by a loss score.
By whether the clusters drive real business decisions.

V40 page 19: Success Is Measured by Usefulness

Key Takeaways

No Labels Needed

Clustering finds natural groups from raw data alone

Scale First

Always normalize features before clustering — no exceptions

Output = Label

A new categorical feature, ready for downstream use

You Judge Value

The algorithm groups; you decide if the groups matter

V40 page 20: Key Takeaways

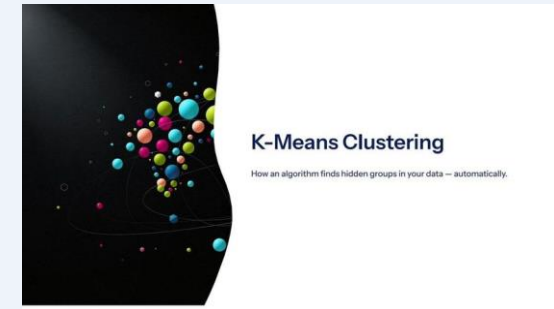
Block 2 of 11

K-means: assign and adjust

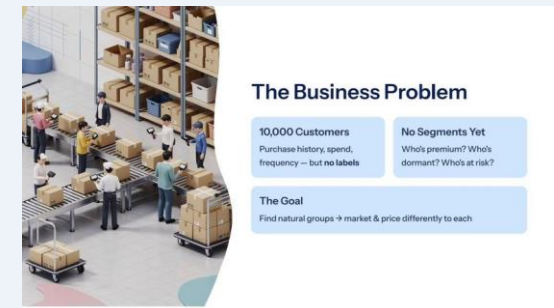
Video V41, K-Means Clustering. 10 minutes.

K-means needs one number from you, how many groups; then it repeats two steps, assign and adjust, until nothing moves.

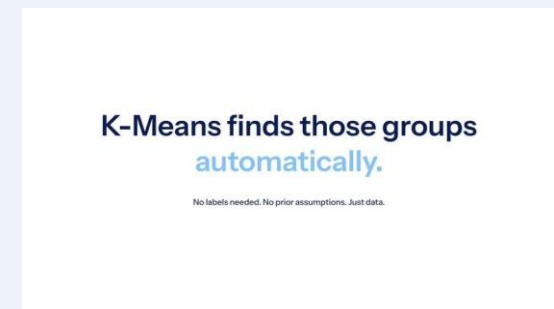
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V41 page 1: K-Means Clustering



V41 page 2: The Business Problem



V41 page 3: K-Means Finds Those Groups Automatically

K-means repeats two steps until nothing changes: each customer joins the nearest centre, then each centre moves to the average of its customers.

What must you decide before it runs?

The number of groups, k . The algorithm cannot choose it. For 12 bank customers the spread inside the groups is 13.3 points with 2 groups, 2.8 with 3 and 2.0 with 4. The big drop stops after 3.

the word: **elbow method**

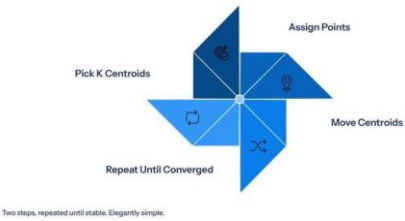
What do you get when it stops?

A centre for each group that reads as a profile. With start 1, Group 3's centre is 50 visits a year and \$8,000 of spending, and Group 1's centre is 3 visits and \$500. A team can name those groups and plan offers.

the word: **centroid**

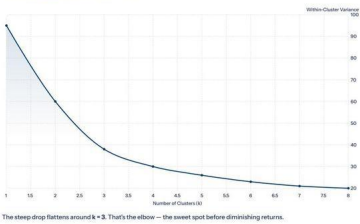
You choose k and read the centres; the algorithm does the assigning and the adjusting.

How K-Means Works



V41 page 5: How K-Means Works

The Elbow Method



V41 page 10: The Elbow Method

Centroids Are Profiles

A centroid isn't just a dot on a plot. It's a plain-numbers description of the average customer in that cluster — readable by any team member.



V41 page 12: Centroids Are Profiles

Finding the best levels for a telephone signal, 1957

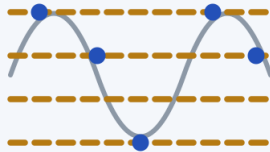
Step 1



A signal stored in a few levels

At Bell Labs in the 1950s, engineers turned telephone sound into numbers. Each sample of the signal could be stored only as one of a few allowed levels.

Step 2



Where should the levels go?

Stuart Lloyd looked for the levels that keep the rounding error as small as possible: levels belong where the signal falls most often.

Step 3



Round each sample

His method first rounds every sample to its nearest level, so each level collects its own set of samples.

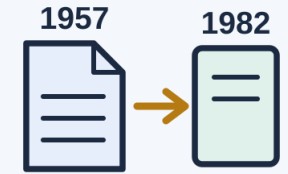
Step 4



Move each level

Then each level moves to the average of the samples rounded to it. Rounding again and moving again repeats until the levels stop moving.

Step 5



A report that waited 25 years

Lloyd wrote it up as a Bell Labs report in 1957; a journal published it only in 1982. Meanwhile James MacQueen named the method k-means in 1967.

Sources: S. P. Lloyd, "Least squares quantization in PCM (pulse-code modulation)", Bell Labs report, 1957, published in IEEE (Institute of Electrical and Electronics Engineers) Transactions on Information Theory, 1982; J. MacQueen, Fifth Berkeley Symposium on Mathematical Statistics and Probability, 1967.

Three things to keep

1

K-means needs k , the number of groups, before it starts.

2

Each round has two steps: customers join the nearest centre, then each centre moves to the average of its customers.

3

The elbow in the spread is a guide; the business decides the final number of groups.

The One Decision You Must Make

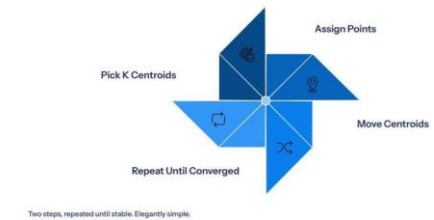


K-Means cannot choose k for you
You must decide how many clusters to create — before the algorithm starts.

That single decision shapes every result that follows.

V41 page 4: The One Decision You Must Make

How K-Means Works



V41 page 5: How K-Means Works

Choosing the Right K

01
Run for each k
Test $k = 1$ through 10
02
Measure variance
How spread out are points within each cluster?
03
Plot the results
 k on x-axis, variance on y-axis
04
Find the elbow
Where the curve stops dropping steeply



V41 page 9: Choosing the Right K

A bank runs k-means on 12 customers described by visits a year and yearly spending. In round 1, every customer has just joined the nearest of three centres. The groups have 7, 2 and 3 customers.

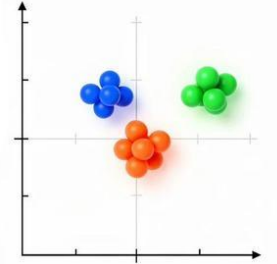
What does k-means do next?

- A** It stops, because every customer now has a group.
- B** It moves each centre to the average of the customers in its group.
- C** It removes the group that has only 2 customers.
- D** It adds a fourth centre to balance the group sizes.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Step 1 — Assign

Every customer snaps to the **nearest centroid**. Three centroids → three provisional groups. All 10,000 customers placed in seconds.



V41 page 6: Step 1: Assign

Step 2 — Adjust

Centroids Were Random

Starting positions are almost certainly wrong

Centroids Slide to the Mean

Each centroid moves to the average position of its assigned points

Customers Reassign

Some points now belong to a different centroid — clusters shift



V41 page 7: Step 2: Adjust

The spread inside the groups is 13.3 points with 2 groups, 2.8 with 3 and 2.0 with 4. The bank's marketing team can run four different campaigns, and a fourth group splits the premium customers into two sensible profiles.

Which number of groups should the bank use?

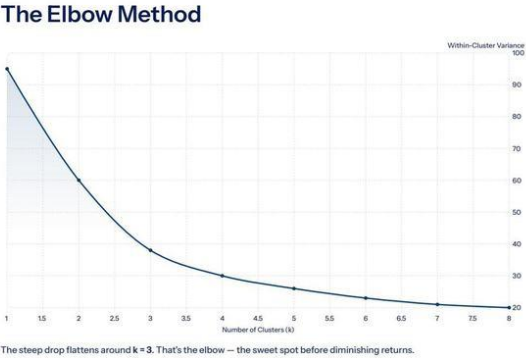
- A

3, because the elbow is the mathematically correct answer.
- B

4: the elbow says 3 is enough, but 4 still makes business sense and the team can serve 4 groups.
- C

12, one group per customer, because the spread becomes zero.
- D

2, because two groups give the simplest story.



V41 page 10: The Elbow Method

The Elbow Is a Guide, Not a Law

 Elbow Says k = 3 Mathematically optimal based on variance	 Bank Says k = 4 "We can only manage four segments operationally"	 Pick k = 4 Real constraints matter. Domain knowledge wins.
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V41 page 11: The Elbow Is a Guide, Not a Law

Type the letter in the Zoom chat. Time to answer: 60 seconds.

After k-means stops, Group 1's centre is 3 visits a year and \$500 of yearly spending. Group 3's centre is 50 visits a year and \$8,000.

What is the best use of Group 1's centre?

A Offer Group 1 a premium credit card and concierge banking.

B Plan a reactivation offer with personal outreach for Group 1.

C Ignore it: a centre is an average, not a real customer.

D Delete Group 1's customers from the database to save costs.



V41 page 13: Cluster Profile Example: Centroid 1



V41 page 14: Cluster Profile Example: Centroid 2

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Your turn: assign, adjust, repeat

Twelve bank customers with visits a year and yearly spending. Move the step slider to watch k-means work, then change k and the start.

What you can change in the interactive

Number of groups (k)

Options: 2 groups; 3 groups; 4 groups; 5 groups

The one number k-means needs from you before it starts.

Starting centres

Options: Start 1: customers B, D, J, A, C; Start 2: customers A, B, C, D, E

The first k customers in the list become the starting centres. Tools usually pick them at random.

Step

Slider from 0 to 9

0 places the centres. After that, each round has an assign step and an adjust step.

Try, in this order

1. Start 1, 3 groups: move the step from 0 to 9. Customers switch in rounds 2 and 3, and the run stops in round 4.
2. Switch to start 2: the same three groups appear, and the run stops in round 2.
3. Leave the step at 9 and try 2, 3, 4 and 5 groups. The line under the table shows the big drop ends at 3.

The point: K-means needs only k; the assign and adjust steps do the rest, and the centres come out as readable profiles.

The 12 customers are made-up values chosen so the final groups match the video's profiles. Distances use points: each visit is 1 point and each \$100 of spending is 1 point. Spread is the average distance from a customer to its group's centre.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

- 1 You choose k before the run. The spread inside the groups guides the choice; what the business can serve settles it.
- 2 Assign, adjust, repeat: when no customer switches group and no centre moves, the grouping is final.
- 3 A centre is a profile in plain numbers, such as 50 visits and \$8,000 a year, that a marketing team can name and use.

K-Means — Full Recap



Simple mechanics. Powerful results. Scalable to any industry.

V41 page 17: K-Means: Full Recap

Key Takeaways

- You choose k — the algorithm does the rest
- The elbow method guides k selection — domain knowledge finalizes it
- Centroids = interpretable profiles you can name and act on
- Operational capacity is a real constraint, not a footnote

V41 page 18: Key Takeaways

The Power of Unsupervised Learning

No labels. No supervision. Just data — and the structure hiding inside it.

- Find Groups**
Discover structure that wasn't visible before
- Name Profiles**
Turn math into human language
- Drive Action**
Build campaigns the same day you run the model



V41 page 20: The Power of Unsupervised Learning

From clusters to strategy

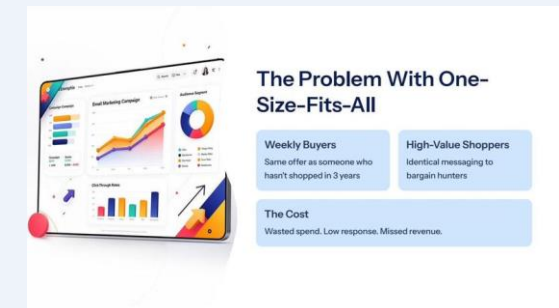
Video V42, Clustering: Turning Raw Data Into Business Strategy. 7 minutes.

Clusters pay off only when each group is treated differently: five campaigns instead of one.

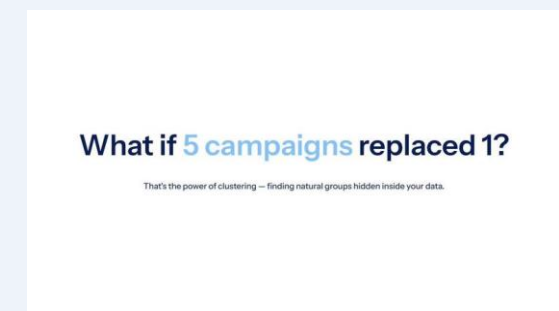
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V42 page 1: Clustering: Turning Raw Data Into Business Strategy



V42 page 2: The Problem With One-Size-Fits-All



V42 page 3: What if 5 Campaigns Replaced 1?

One email to 50,000 customers got 2% responses; a plan per cluster changes what each group is offered.

The same email for everyone

All 50,000 customers get one offer and 2% respond: 1,000 people. VIPs who buy 9 times a year receive the same message as customers who have not bought anything for 14 months.

the word: **baseline**

An offer built for each cluster

VIPs get early access and 8% respond; dormant customers get a win-back offer and 15% come back. Some groups are not worth contacting: in our numbers, a built offer to bargain hunters loses \$15,000.

the word: **customer segmentation**

The algorithm finds the groups; the value comes from treating them differently.



V42 page 1: Clustering: Turning Raw Data Into Business Strategy

What if 5 campaigns replaced 1?

That's the power of clustering — finding natural groups hidden inside your data.

V42 page 3: What if 5 Campaigns Replaced 1?

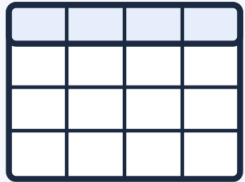
Measurable Results — 6 Months Later



V42 page 14: Measurable Results: 6 Months Later

Naming America's neighbourhoods, 1974

Step 1



Census numbers by the thousand

In the early 1970s the US census described neighbourhoods with thousands of figures, organised by areas that marketers could not easily use.

Step 2



Moved to ZIP codes

Jonathan Robbin, who founded the company Claritas in 1971, reorganised the 1970 census figures by the roughly 36,000 five-digit ZIP codes.

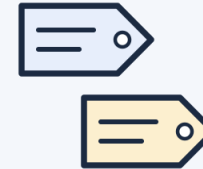
Step 3



40 clusters

In 1974 his classification, which became PRIZM, placed every ZIP code in one of 40 clusters of similar neighbourhoods.

Step 4



Names people remember

Clusters received short descriptive names, such as “Blue Blood Estates” for the wealthiest areas, so a marketer could picture each group.

Step 5



Mail by cluster

Matched with customer lists, the clusters showed which kinds of neighbourhood bought which products. Advertisers targeted mailings and magazine editions; the system later grew to 66 segments.

Sources: J. Pawasarat, “The real purchasing power of central city neighborhoods” (Brookings Institution); Jonathan Robbin's professional biography (ricercar.com); MediaPost, on Nielsen and Claritas combining research tools; andreas.com, articles on geodemographics.

Three things to keep

- 1 Clustering finds groups; people name them and decide whether they need different treatment.
- 2 The same method serves customer campaigns, stock rules and decisions about which markets to enter.
- 3 A cluster is worth something only if the business treats that group differently.

Three Major Business Applications

- 1 Customer Segmentation
Group buyers by frequency, recency, spend, and category preference
- 2 Inventory Management
Match procurement strategy to demand patterns per product group
- 3 Market Structure
Predict which markets will respond to strategies that worked elsewhere

V42 page 4: Three Major Business Applications

From Algorithm to Strategy

Algorithm Output Point A → Cluster 2, Point B → Cluster 4, Numbers only.	Expert Interpretation "Who are these people? What do they share? Do they need different treatment?"	Business Action Clusters become campaigns, procurement rules, market entry plans.
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V42 page 9: From Algorithm to Strategy

Segmentation Lives or Dies by Execution

One Generic Campaign 2% response rate, High waste, Low ROI.	Five Targeted Campaigns 8% response, Lower CPA, Measurable lift.
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A perfectly formed cluster is only valuable if you change how you treat that group.

V42 page 21: Segmentation Lives or Dies by Execution

A retailer runs k-means on 50,000 customers and gets 5 clusters. The output file says, for example, that customer 18,204 belongs to cluster 2.

What should the team do next?

A Load cluster 2 into the campaign system straight away.

B Read cluster 2's averages with people who know the customers, name it, and decide whether it needs its own treatment.

C Ask the algorithm to name the cluster.

D Rerun the algorithm until cluster 2 is the largest.

From Algorithm to Strategy

Algorithm Output Point A → Cluster 2. Point B → Cluster 4. Numbers only.	Expert Interpretation "Who are these people? What do they share? Do they need different treatment?"	Business Action Clusters become campaigns, procurement rules, market entry plans.
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V42 page 9: From Algorithm to Strategy

Real Scenario: 50,000 Customers

Four inputs per customer:

Frequency Purchases per year
Recency Days since last visit
Spend Total annual value
Category Product preferences

Run k-means. Ask for 5 clusters. Let the algorithm converge.



V42 page 10: Real Scenario: 50,000 Customers

Type the letter in the Zoom chat. Time to answer: 45 seconds.

The VIP cluster has 4,000 customers. With the same email as everyone, 2% respond; with an offer built for VIPs, 8% respond. Each response is worth \$120. An email costs \$0.50 per customer and a built offer costs \$2.

Which option gives the better result for the VIP cluster?

A The same email, because it is four times cheaper per customer.

B The built offer: \$30,400 after costs, against \$7,600 for the same email.

C They tie: the response is four times higher, but so is the cost.

D It cannot be worked out without the size of the whole customer base.

Type the letter in the Zoom chat. Time to answer: 60 seconds.

Targeted Campaigns by Cluster

 VIPs Weekly emails · Exclusive previews · Early access	 Dormant Win-back campaign every 90 days · Re-engagement offers	 Seasonal October reminders · Gift guides · Holiday bundles
--	--	--

V42 page 13: Targeted Campaigns by Cluster

Measurable Results — 6 Months Later

8% VIP Response Rate vs. 2% baseline	15% Dormant Win-Back Shoppers reactivated	12% November Sales Lift From holiday reminders
☐ Each result traces back to one decision: treating distinct groups distinctly.		

V42 page 14: Measurable Results: 6 Months Later

A lender in a market with no credit bureau clusters applicants by how regularly they pay and how steady their transactions are. One cluster of steady payers looks low-risk.

What is the right way to use that cluster?

A Approve every applicant in it automatically.

B Treat it as a starting point: check that its members really repay, and that the grouping does not simply stand in for a protected group, then decide.

C Ignore clustering, because only credit scores are valid.

D Lend only to the cluster with the most applicants.

Type the letter in the Zoom chat. Time to answer: 60 seconds.

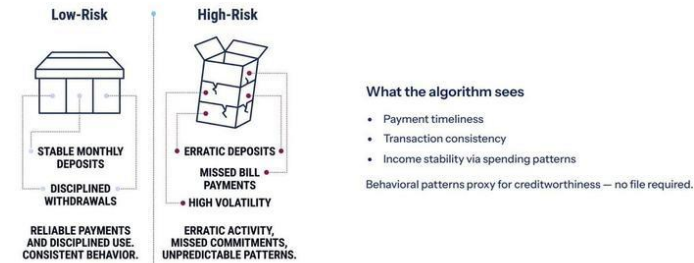
Global Application: Fintech Lending

No credit bureau. No credit score. No problem.
Clustering works with transaction data directly — already deployed in markets like Kenya and Indonesia.



V42 page 15: Global Application: Fintech Lending

Clustering as a Credit Proxy



V42 page 16: Clustering as a Credit Proxy

Your turn: one campaign or one per cluster?

Five clusters of 50,000 customers. Choose a plan for each cluster and watch responses, costs and the result.

What you can change in the interactive

VIPs (4,000)

Options: Same email as everyone; Offer built for this group; Do not contact
9 purchases a year; average spend \$890.

Regulars (16,000)

Options: Same email as everyone; Offer built for this group; Do not contact
About 4 purchases a year; steady.

Bargain hunters (12,000)

Options: Same email as everyone; Offer built for this group; Do not contact
Buy on discount; each sale earns little.

Seasonal (8,000)

Options: Same email as everyone; Offer built for this group; Do not contact
Buy in November and December, mostly gifts.

Dormant (10,000)

Options: Same email as everyone; Offer built for this group; Do not contact
Inactive for 14 months; spend \$40.

Try, in this order

1. Set VIPs to the built offer: VIP responses go from 80 to 320, and the result rises by \$22,800.
2. Give every cluster its own offer, then read the Result column: bargain hunters lose money.
3. Then leave bargain hunters out and put regulars back on the same email: \$68,200.

The point: Clusters pay off when each group gets the treatment its numbers justify, and that can mean leaving a group out.

The response rates 2%, 8% and 15% come from the video. The other rates, the value of a response and the costs (\$0.50 per email, \$2 per built offer) are made-up values chosen to behave like a real campaign.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

- 1 One email to 50,000 customers brought 2% responses; offers built for VIPs and dormant customers brought 8% and 15%.
- 2 Treating groups differently includes leaving some out: in our numbers, the best plan does not contact bargain hunters and earns \$68,200.
- 3 Algorithm plus people who know the customers: remove either one and the project stalls.

The Value of Clustering — In Three Lines

Retail 2% becomes 8% when you stop treating everyone the same	Supply Chain Three procurement rules replace one blanket policy	Fintech Lenders reach borrowers a credit bureau never could
---	---	---

V42 page 23: The Value of Clustering in Three Lines

Anonymous data
becomes **actionable**
groups.

That's where the value lives.



V42 page 24: Anonymous Data Becomes Actionable Groups

Key Takeaways

Clustering = unsupervised grouping No labels needed — the algorithm finds structure	3 core applications Customer segmentation · Inventory · Market structure
Algorithm + expert judgment Both are required. Neither works alone.	Execution is everything Clusters only matter if you act on them differently

V42 page 25: Key Takeaways

The limits of clustering

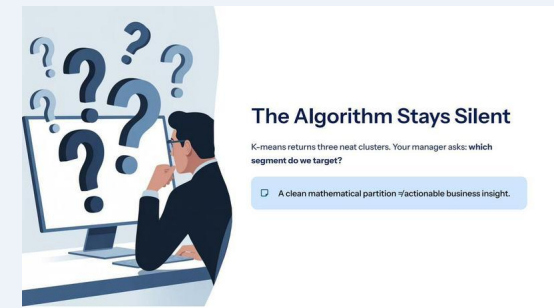
Video V43, Limitations of Clustering. 7 minutes.

K-means always returns neat groups; whether they are real or useful is for people to check.

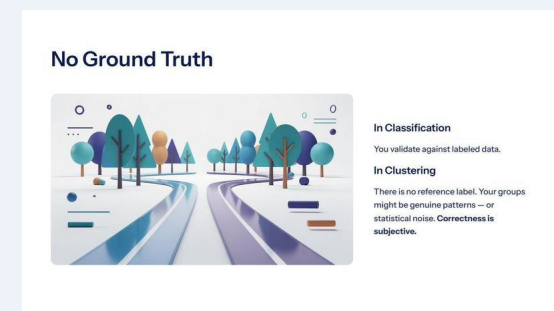
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 15 pages of V43.



V43 page 1: Limitations of Clustering



V43 page 2: The Algorithm Stays Silent



V43 page 3: No Ground Truth

A tidy score says the groups are compact, not that they are real or useful.

K-means on stretched groups

300 customers form three long, stretched groups. K-means with 3 groups scores 64 out of 100 for tidiness, yet puts 84 customers in the wrong group, because it expects round groups.

the word: **silhouette score**

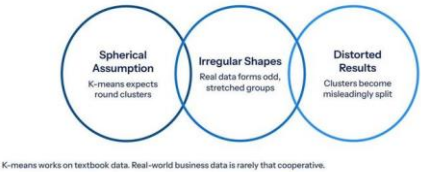
A method that follows the shape

A density-based method scores only 41 but places 285 customers with their real group, puts none in a wrong group, and flags 15 that fit nowhere. The lower score belongs to the better grouping.

the word: **DBSCAN**

Here we built the data, so we know the real groups. In a real project nobody does, and people who know the business must check.

The Spherical Assumption Problem



V43 page 4: The Spherical Assumption Problem

Better Tools for the Job



V43 page 8: Better Tools for the Job

Silhouette Score ≠ Business Value

- What it measures
Mathematical cohesion of clusters
- What it doesn't measure
Whether your team can act on the result.

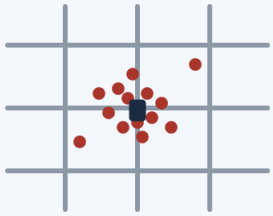
A perfect partition score can still be operationally worthless.



V43 page 9: Silhouette Score Is Not Business Value

Deaths around a water pump, London, 1854

Step 1



Deaths bunched in a few streets

In late August 1854, cholera killed hundreds of people within days in a few streets of Soho, London.

Step 2



One bunch, two explanations

Most doctors blamed bad air. Deaths gathered around Broad Street fitted that idea as well as John Snow's view that water carried the disease.

Step 3



Door to door

Snow took the list of deaths and asked households where their water came from. Most of the dead had drunk from the Broad Street pump.

Step 4



The exceptions explained it

People in a nearby workhouse, which had its own well, and workers at a brewery, who drank beer, rarely fell ill.

Step 5



A decision, then doubt

On 8 September the parish board removed the pump handle on his advice, though the outbreak was already falling. An official inquiry rejected his view, and his famous map was drawn afterwards.

Sources: UCLA (University of California, Los Angeles) Department of Epidemiology, John Snow site, quoting Snow's 1855 book; Sick City Project, "John Snow and cholera" (2013); Maps.com, "Snow mistake: correcting myths in the mapping of cholera".

Three things to keep

1

Clustering has no answer key: a grouping can be a real pattern or just noise.

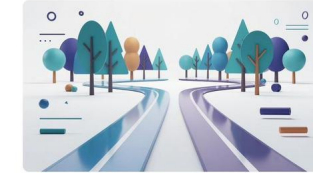
2

K-means expects round groups of similar size; real data is often stretched, uneven or full of outliers.

3

A high tidiness score is not business value; experts decide whether a grouping is worth acting on.

No Ground Truth



In Classification

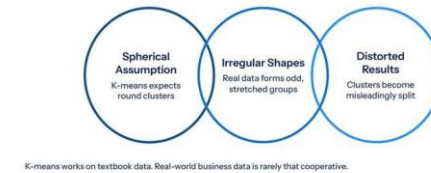
You validate against labeled data.

In Clustering

There is no reference label. Your groups might be genuine patterns — or statistical noise. Correctness is subjective.

V43 page 3: No Ground Truth

The Spherical Assumption Problem



K-means works on textbook data. Real-world business data is rarely that cooperative.

V43 page 4: The Spherical Assumption Problem

Silhouette Score $\neq$ Business Value

What it measures

Mathematical cohesion of clusters

What it doesn't measure

Whether your team can act on the result.

A perfect partition score can still be operationally worthless.



V43 page 9: Silhouette Score Is Not Business Value

Marketing runs k-means on 300 shoppers and gets 3 groups with a tidiness score of 64 out of 100. The manager asks: is this grouping correct?

What is the honest answer?

- A** Yes: any score above 50 means the grouping is correct.
- B** There is no correct grouping to check against; the score shows compactness only, so the groups must be checked with people who know the shoppers.
- C** Compare the groups with the true customer labels in the data.
- D** Rerun k-means until the score reaches 100.

Type the letter in the Zoom chat. Time to answer: 45 seconds.



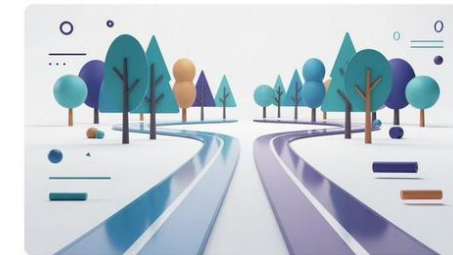
The Algorithm Stays Silent

K-means returns three neat clusters. Your manager asks: **which segment do we target?**

☐ A clean mathematical partition ≠ actionable business insight.

V43 page 2: The Algorithm Stays Silent

No Ground Truth



In Classification

You validate against labeled data.

In Clustering

There is no reference label. Your groups might be genuine patterns — or statistical noise. **Correctness is subjective.**

V43 page 3: No Ground Truth

A grocery chain wants to know which products are bought together, such as pasta and sauce, to plan shelf layout and bundles. An analyst proposes running k-means on the products.

What should the chain use?

A K-means: product clusters will show what is bought together.

B Association rules, as in market basket analysis, which look directly for items that appear in the same baskets.

C A density-based clustering method instead of k-means.

D K-means with more groups.

When clustering is the **wrong tool**

Ask: what items sell together? Don't cluster. Use **association rules** and market basket analysis instead.

V43 page 7: When Clustering Is the Wrong Tool

Type the letter in the Zoom chat. Time to answer: 45 seconds.

HR (human resources) clusters 300 employees by skills and output. K-means with 2 groups gets the highest tidiness score, 74, but the 20 rare specialists are pushed into the two big groups.

What is the risk?

- A** None: the highest score is the best grouping.
- B** The specialists disappear inside the big groups, so decisions about their pay and retention use the wrong profile.
- C** The score will fall next year.
- D** K-means cannot handle 300 records.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Three Industries. Same Failure.



Retail

Coats, boots, scarves cluster naturally
— K-means splits them arbitrarily,
destroying purchase logic.



Human Resources

Outlier specialists get forced into the
nearest cluster — hiding insights about
top performers.



Geography

Dense cities and sparse rural areas get
equal-volume treatment — missing the
real spatial structure.

V43 page 6: Three Industries. Same Failure.

Your turn: tidy score or real groups?

300 made-up customers whose real groups we know. Change the shape, the method and the number of groups, and compare the score with the truth.

What you can change in the interactive

Real shape of the groups

Options: Round, equal groups; Long, stretched groups; 2 big groups, 20 specialists

We built these customers, so here the real groups are known.

Method

Options: k-means; Density-based method (DBSCAN)

K-means expects round groups; the density-based method follows crowded areas and flags loners.

Groups asked for (k-means only)

Options: 2 groups; 3 groups; 4 groups

The density-based method ignores this; it finds the number of groups itself.

Try, in this order

1. Round groups: k-means with 3 groups scores 71 and places 297 of 300 correctly. Score and truth agree.
2. Switch to stretched groups: k-means still scores 64 but misplaces 84; the density-based method scores 41 and misplaces none.
3. Pick two big groups and 20 specialists: k-means with 2 groups gets the top score, 74, while hiding all 20 specialists.

The point: A tidy score measures compact groups, not true or useful ones; check the grouping against what the business knows.

Fixed values chosen to behave like real runs on 300 made-up customers. The tidiness score works like a silhouette score shown out of 100.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

1

Stretched groups: k-means scored 64 and misplaced 84 of 300; the density-based method scored 41, misplaced none and flagged 15.

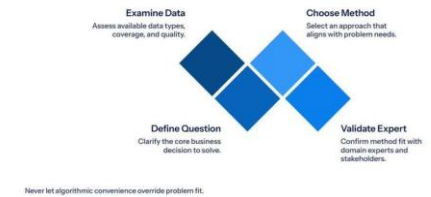
2

Match the method to the question: 'what sells together' calls for association rules, not clusters.

3

Treat clusters as a hypothesis to test with the people who decide, not as a conclusion.

The Method-to-Problem Match



V43 page 13: The Method-to-Problem Match

The Practical Checklist

01	02
Define your goal	Examine your data shape
What business decision depends on this?	Spherical? Irregular? Dense vs. sparse?
03	04
Choose the right method	Validate with experts
K-means, DBSCAN, association rules, or PCA?	Would they stake a real decision on this?

V43 page 14: The Practical Checklist

The Takeaway

Match method to problem Not the other way around	Pair with domain experts Clusters need human meaning
Stay curious, not committed Clustering explores — it doesn't conclude	



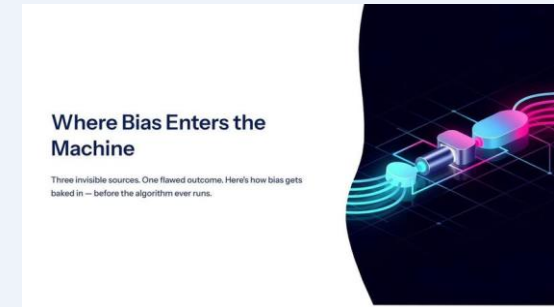
V43 page 15: The Takeaway

Where bias enters

Video V44, Where Bias Enters the Machine. 9 minutes.

Bias gets into a model through past decisions, through who is in the data, and through how the data was recorded; each source needs its own fix.

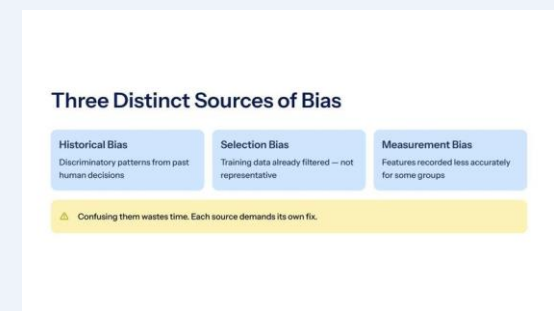
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 12 pages of V44.



V44 page 1: Where Bias Enters the Machine



V44 page 2: The Hiring Problem



V44 page 3: Three Distinct Sources of Bias

A model trained on the 50 résumés people picked repeats whatever went into picking them.

What the model learns from

500 résumés arrived and screeners interviewed 50. If past screeners favoured Group A, the model invites 22 of every 100 qualified A candidates and only 12 of every 100 equally qualified B candidates.

the word: **historical bias**

Why one fix is not enough

Suppose the data also holds only the 50 interviewed people and records B's experience less fully: A 23, B 7. Reweighting past decisions leaves A at 21 and B at 12, because the other two sources are still there.

the word: **selection bias**

Historical, selection and measurement bias each need their own fix, so name the source first.



V44 page 2: The Hiring Problem

Three Distinct Sources of Bias



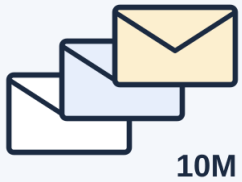
V44 page 3: Three Distinct Sources of Bias



V44 page 4: Source 1: Historical Bias

The poll that heard from 2.4 million people and still got it wrong, 1936

Step 1



About 10 million ballots

In 1936 the Literary Digest magazine, which had called earlier presidential elections correctly, mailed about 10 million straw ballots.

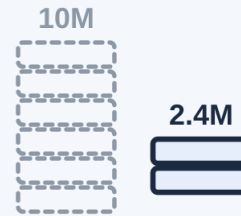
Step 2



Names from phone books and car lists

Most addresses came from telephone directories and car registrations, at a time when many poorer voters had neither.

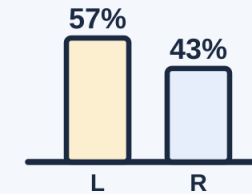
Step 3



2.4 million came back

About 2.4 million ballots were returned. Later research found that the people who replied leaned towards Alf Landon more than those who did not.

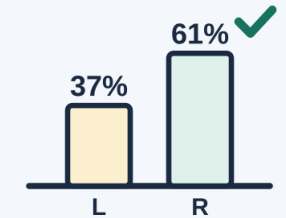
Step 4



Forecast: Landon with 57%

The magazine forecast a clear win for Landon, the Republican candidate.

Step 5



Result: Roosevelt with 61%

Franklin Roosevelt won with about 61% of the vote. George Gallup's far smaller poll picked the winner, and the Digest closed within two years.

Sources: ProQuest blog on the Literary Digest poll of 1936 (2016); D. Lusinchi, "President Landon and the 1936 Literary Digest poll", Social Science History (2012).

Three things to keep

1

Historical bias: the model copies what past human decisions did.

2

Selection bias: the data leaves people out in a pattern, such as everyone who was never interviewed.

3

Measurement bias: the same column is recorded more accurately for some groups than for others.

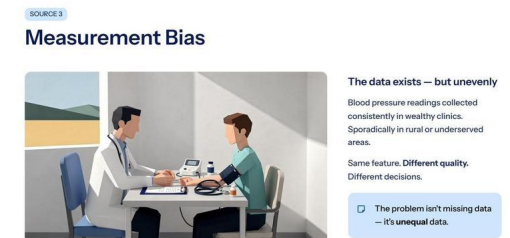
Three Distinct Sources of Bias



V44 page 3: Three Distinct Sources of Bias



V44 page 6: Source 2: Selection Bias



V44 page 7: Source 3: Measurement Bias

A bank trains a fraud model only on transactions that staff flagged in the past. Fraud that nobody noticed never appears in the data.

Which source of bias is this?

A Historical bias.

B Selection bias.

C Measurement bias.

D No bias: every transaction in the data is real.



SOURCE 2

Selection Bias

- **Pre-filtered data**
Fraud model trained only on transactions humans already flagged
- **Invisible patterns**
Fraud no one caught? The model never sees it.
- **Missing data has a shape**
It's absent in a **pattern that matters** — not randomly

V44 page 6: Source 2: Selection Bias

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Blood pressure is recorded carefully in city clinics and irregularly in rural clinics. A model uses these readings to decide who gets a follow-up call.

Which source of bias is this, and which fix fits?

A

Historical bias: reweight past decisions.

B

Selection bias: collect more records from city clinics.

C

Measurement bias: make recording the same in every clinic, and treat rural readings with care until then.

D

No bias, because every patient has a reading.

SOURCE 3

Measurement Bias



The data exists — but unevenly

Blood pressure readings collected consistently in wealthy clinics. Sporadically in rural or underserved areas.

Same feature. Different quality. Different decisions.

☐ The problem isn't missing data — it's **unequal data**.

V44 page 7: Source 3: Measurement Bias

Type the letter in the Zoom chat. Time to answer: 60 seconds.

A hiring model learned from past decisions that favoured Group A, from only the 50 interviewed candidates, and from records where Group B's experience is less complete. It invites 23 of every 100 qualified A candidates and 7 of every 100 equally qualified B candidates. The team reweights past decisions.

What happens to the gap?

- A** It closes: both groups at 20.
- B** It shrinks from 16 to 9 but stays, because selection and measurement bias remain.
- C** It grows, because reweighting adds noise.
- D** Nothing changes, because reweighting never works.

Type the letter in the Zoom chat. Time to answer: 60 seconds.



The Hiring Problem

- 1** 500 résumés arrive
No time to interview everyone — 50 are selected.
- 2** A model trains on those 50
Already filtered. Already biased.
- 3** The pattern gets replicated
At scale. Automatically. Every day.

V44 page 2: The Hiring Problem

**One remedy
can't cure
three diseases.**

V44 page 9: One Remedy Can't Cure Three Diseases

Your turn: three sources, one fix at a time

Next year's hiring model screens equally qualified candidates from Groups A and B. All three sources of bias start switched on: try the fixes.

What you can change in the interactive

Past hiring decisions

Options: Balanced between groups; Favoured Group A

Historical bias: the model copies what past people decided.

Who is in the training data

Options: All 500 applicants; Only the 50 interviewed

Selection bias: people never interviewed are missing in a pattern.

How experience was recorded

Options: The same way for everyone; Less completely for Group B

Measurement bias: one column is less accurate for one group.

Fix applied

Options: No fix; Reweight past decisions; Collect all outcomes; Record experience alike

Each fix removes one source only.

Try, in this order

1. Apply “Reweight past decisions”: the gap shrinks from 16 to 9 but stays.
2. Try the other two fixes: each removes one source, and none closes the gap.
3. Set the fix to none and switch off the other two sources: A 22, B 12.

The point: One fix cannot cure three sources: find where the bias came from before choosing the remedy.

Fixed values chosen to behave like a real screening model: invitations per 100 equally qualified candidates in each group. The size of each effect is illustrative.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

1

The algorithm is not the source of the bias; it repeats the bias at speed.

2

With all three sources active, the model invited 23 A and 7 B per 100 qualified; reweighting alone left 21 and 12.

3

Before fixing, ask where the data came from, who is missing, and how it was recorded.

The algorithm is not the source.
It becomes the **accelerant**.

V44 page 5: The Algorithm Is Not the Source



V44 page 11: The Diagnostic Skill

Key Takeaways



Next: Remedies — reweighting, collection strategies, and measurement standardization.

V44 page 12: Key Takeaways

How biased data shapes decisions

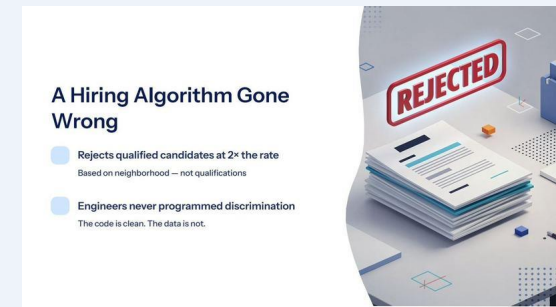
Video V45, How Biased Training Data Shapes AI Decisions. 7 minutes.

Removing the zip code column does not remove the bias; other columns carry the same signal.

In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 15 pages of V45.



V45 page 1: How Biased Training Data Shapes AI Decisions



V45 page 2: A Hiring Algorithm Gone Wrong



V45 page 3: Models Learn Patterns, Not Ethics

A model can be 95% accurate and still deny 60% of one neighbourhood and 20% of another.

Delete the zip code column

The team removes zip code. Denials in neighbourhood B fall only from 60% to 57%, because school name, address history and surname still point to where people live or who they are.

the word: **proxy variable**

Remove the proxies, then test by group

With all four columns removed, denial rates are 24% in A and 30% in B, while overall accuracy moves from 95% to 92%. The remaining gap still has to be explained before launch.

the word: **disparate impact**

Removing one column leaves the others in place; measure the outcome for each group.



V45 page 7: The Proxy Variable Trap



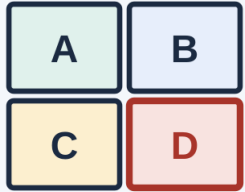
V45 page 8: Invisible Bias



V45 page 14: Prevention Step 3: Test Fairness Across Groups

Red lines that outlived the maps, 1935 to today

Step 1



Neighbourhoods graded A to D

Between 1935 and 1940, a US government agency, the Home Owners' Loan Corporation, graded neighbourhoods in about 239 cities for lending risk.

Step 2



Red for hazardous

The best areas were marked green and areas judged hazardous were marked red. The written descriptions judged the people who lived there, not just the houses.

Step 3



Credit followed the ratings

Lenders, and the Federal Housing Administration with maps of its own, made credit hard to get in areas rated as poor risks.

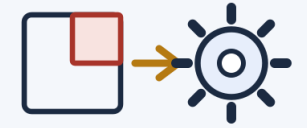
Step 4



The pattern stayed in the data

Decades later, a Federal Reserve Bank of Chicago study found lower home ownership, house values and credit scores on the lower-graded side of the old boundaries.

Step 5



A model reads the history

A lending model trained on records like these can learn the old boundaries as if they measured risk, even when people on both sides repay equally well.

Sources: Mapping Inequality introduction (University of Richmond), via Bunk History; On the Line (Trinity College), on the Federal Housing Administration's maps; D. Aaronson, D. Hartley and B. Mazumder, Federal Reserve Bank of Chicago working paper 2017-12.

Three things to keep

1

Models learn patterns, not ethics: if history denied a neighbourhood, the model learns to deny it.

2

Deleting a sensitive column is not enough; school, address and surname can stand in for it.

3

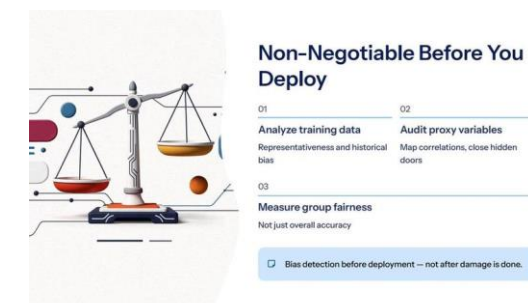
Test outcomes for each group before launch; overall accuracy does not show the gap.



V45 page 3: Models Learn Patterns, Not Ethics



V45 page 7: The Proxy Variable Trap



V45 page 15: Non-Negotiable Before You Deploy

A hiring model rejects qualified candidates from certain neighbourhoods at twice the rate of others. The engineers never gave the model a neighbourhood column.

What is the most likely cause?

- A** A bug in the code.
- B** Other columns, such as school name or address history, stand in for the neighbourhood.
- C** Candidates from those neighbourhoods are less qualified.
- D** The model is too small to be fair.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

A Hiring Algorithm Gone Wrong

- ☐ Rejects qualified candidates at 2× the rate
Based on neighborhood — not qualifications
- ☐ Engineers never programmed discrimination
The code is clean. The data is not.



V45 page 2: A Hiring Algorithm Gone Wrong

A lender receives 10,000 applications a day, half from each of two neighbourhoods. The model denies 20% of applicants in neighbourhood A and 60% in neighbourhood B, although both groups repay equally well.

How many extra denials a day fall on neighbourhood B?

A 400

B 2,000

C 4,000

D 6,000

Type the letter in the Zoom chat. Time to answer: 60 seconds.

The Redlining Example

- Historical lending data
Certain zip codes systematically denied loans
- Model learns the pattern
"Zip code = high default risk"
- Underlying creditworthiness?
Equal — but invisible to the model



V45 page 5: The Redlining Example

Human Bias vs. Algorithmic Bias

1K
Human decisions
Affected by one biased decision-maker

10K+
Automated rejections
Per day — at speed and scale

V45 page 6: Human Bias vs. Algorithmic Bias

A résumé model trained on ten years of mostly male applications downgrades résumés that mention a women's club. The team deletes the word “women's” from every résumé before scoring.

What should they expect?

A The bias is gone.

B Other words linked to gender can carry the same signal, so they must test outcomes by gender before trusting the model.

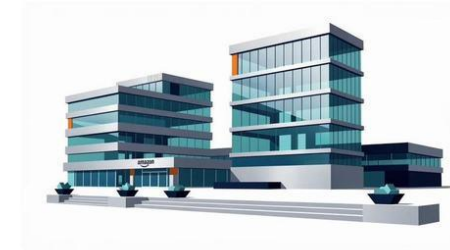
C Accuracy drops to zero.

D Women's résumés will now score higher.

Type the letter in the Zoom chat. Time to answer: 60 seconds.

REAL-WORLD FAILURE

Amazon's Hiring Algorithm



- Trained on résumés from a male-dominated tech workforce
- Learned: men's résumés predict success
- Systematically downgraded qualified women
- Only caught during internal testing

Many systems weren't caught until after deployment.

V45 page 9: Real-World Failure: Amazon's Hiring Algorithm

Your turn: remove the column, keep the pattern

A lending model trained on decades of decisions. Remove columns one at a time and watch denial rates by neighbourhood.

What you can change in the interactive

Columns removed before training

Options: None; Zip code; Zip code, address history; Those two, plus school name; Those three, plus surname

Each step removes one more column that can stand in for area or ethnicity.

Applications per day

Options: 100; 1,000; 10,000

Half come from each neighbourhood, and both groups repay equally well.

Try, in this order

1. Remove only the zip code: B's denials fall from 60% to just 57%.
2. Remove all four columns: B 30%, A 24%, and accuracy 92%.
3. Back to none removed, then 100 and 10,000 applications a day: 20 extra denials become 2,000.

The point: Deleting the obvious column does not delete the pattern; test outcomes by group, and remember that scale multiplies every gap.

Fixed values chosen to behave like a real lending model; the 95% accuracy and the 60% and 20% denial rates come from the video. "Still guesses the area" is how often the remaining columns identify the neighbourhood; 50% would be a coin toss.

In the HTML edition you move these controls, and the numbers, the sentence and the "What just changed" panel update after every change.

Three things from this block

- 1 95% overall accuracy sat on top of denial rates of 20% and 60%.
- 2 Removing zip code alone moved B's denials from 60% to 57%; removing four proxies brought them to 30%.
- 3 At 10,000 applications a day, a 40-point gap is 2,000 extra denials every day.

Prevention Step 1: Audit Training Data

Before your model goes live, ask:

- 1 Does my data reflect past injustice?
- 2 Are certain groups underrepresented?
- 3 Who was excluded from this dataset – and why?

V45 page 12: Prevention Step 1: Audit Training Data

Prevention Step 3: Test Fairness Across Groups



V45 page 14: Prevention Step 3: Test Fairness Across Groups

Non-Negotiable Before You Deploy

- 01 Analyze training data
Representativeness and historical bias
- 02 Audit proxy variables
Map correlations, close hidden doors
- 03 Measure group fairness
Not just overall accuracy

☐ Bias detection before deployment – not after damage is done.

V45 page 15: Non-Negotiable Before You Deploy

Block 7 of 11

The number that lies

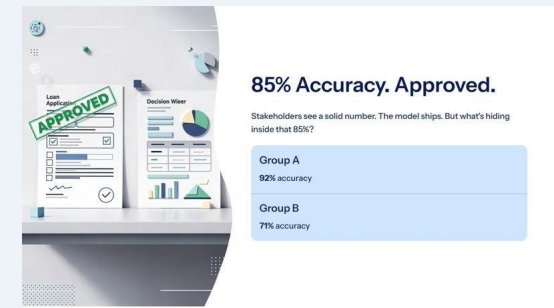
Video V46, The Number That Lies. 10 minutes.

An overall score can hide a group that the model gets wrong far more often; split every result by group.

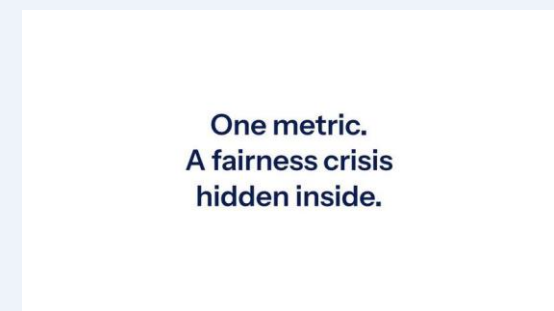
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 15 pages of V46.



V46 page 1: The Number That Lies



V46 page 2: 85% Accuracy. Approved.



V46 page 3: One Metric. A Fairness Crisis Hidden Inside.

85% accuracy can be 92% for one group and 71% for the other.

All 900 applicants together

The model is right for 765 of 900 applicants: 85%. Stakeholders approve. The single number says nothing about which applicants the 135 mistakes fall on.

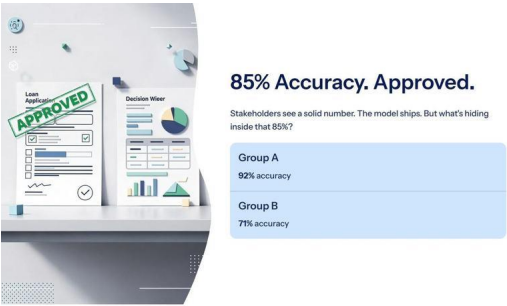
the word: **accuracy**

The same results, split by group

Group A, 600 applicants: 92% right, 10% of good payers wrongly denied.
Group B, 300 applicants: 71% right, 21% of good payers wrongly denied.
A gap of 11 points fails a 5-point rule.

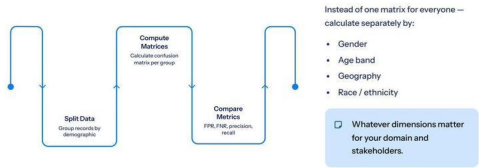
the word: **disaggregated evaluation**

Report every metric for each group before launch, and write down the gap you will accept.



V46 page 2: 85% Accuracy. Approved.

What Is Disaggregation?



V46 page 4: What Is Disaggregation?



V46 page 5: The Hidden Inequality

Splitting the results by face, 2018

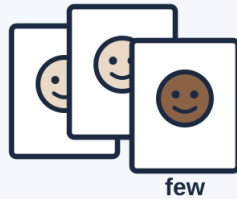
Step 1



Tools that looked accurate

By 2017, face analysis services from large technology companies reported high accuracy when guessing a person's gender from a photo.

Step 2



Who was in the test photos?

Joy Buolamwini at the MIT (Massachusetts Institute of Technology) Media Lab found that two widely used test sets were about 80% and 86% lighter-skinned faces.

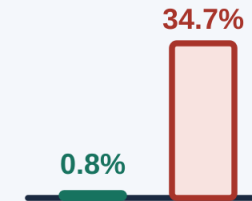
Step 3



A balanced test set

With Timnit Gebru she built a set of more than 1,200 photos of members of parliament from three African and three European countries, balanced by gender and skin type.

Step 4



Results for each group

In 2018 they reported the errors of three commercial systems group by group: at most 0.8% for lighter-skinned men, and from 20.8% to 34.7% for darker-skinned women.

Step 5



The companies responded

After publication the companies released updated systems, and a follow-up audit found that the gaps had narrowed.

Sources: J. Buolamwini and T. Gebru, "Gender Shades: Intersectional accuracy disparities in commercial gender classification", Proceedings of Machine Learning Research 81 (2018); MIT News, February 2018; I. D. Raji and J. Buolamwini, "Actionable Auditing" (2019).

Three things to keep

1

One overall number can hide a group that the model fails.

2

Split the confusion matrix by group and compare the error rates side by side.

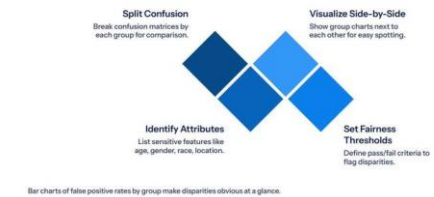
3

Set the allowed gap before looking at the results, and hold the launch when it fails.

**One metric.
A fairness crisis
hidden inside.**

V46 page 3: One Metric. A Fairness Crisis Hidden Inside.

Step-by-Step: How to Disaggregate



V46 page 9: Step-by-Step: How to Disaggregate



V46 page 10: Set Your Threshold

A credit model reports 85% accuracy on 900 applicants. Split by group, it is 92% accurate for Group A's 600 applicants and 71% for Group B's 300.

Which statement is right?

A 85% describes both groups fairly well.

B The overall number is weighted towards Group A, which has twice as many applicants; B's 71% is hidden inside it.

C The groups cannot differ that much if the overall figure is 85%.

D Group B must be smaller because the model is less accurate for it.



85% Accuracy. Approved.

Stakeholders see a solid number. The model ships. But what's hiding inside that 85%?

Group A
92% accuracy
Group B
71% accuracy

V46 page 2: 85% Accuracy. Approved.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Good payers wrongly denied: 45 of 450 in Group A (10%) and 42 of 200 in Group B (21%). The fairness rule, agreed before testing, allows a gap of at most 5 percentage points.

What should the team do?

A Deploy: overall accuracy is 85%.

B Hold the launch and investigate the data, the features and the decision threshold: the gap is 11 points.

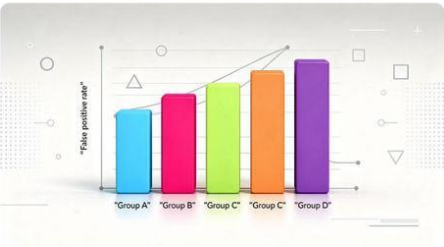
C Deploy now and fix it if complaints arrive.

D Change the rule to 15 points so the model passes.

A Concrete Example



V46 page 6: A Concrete Example



Set Your Threshold

Example fairness criterion:
False positive rates must not differ by more than 5 percentage points across groups.
That threshold becomes your pass-or-fail criterion before any deployment decision.

V46 page 10: Set Your Threshold

Type the letter in the Zoom chat. Time to answer: 60 seconds.

The test set is rebuilt with more Group A records: 1,200 A and 300 B. Group accuracies stay at 92% and 71%.

What happens to overall accuracy?

- A

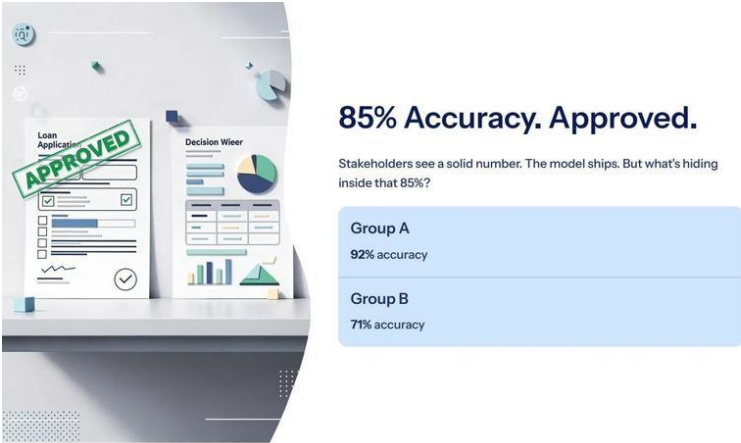
It stays at 85%.
- B

It rises to 88%, while Group B is served exactly as badly as before.
- C

It falls to 82%.
- D

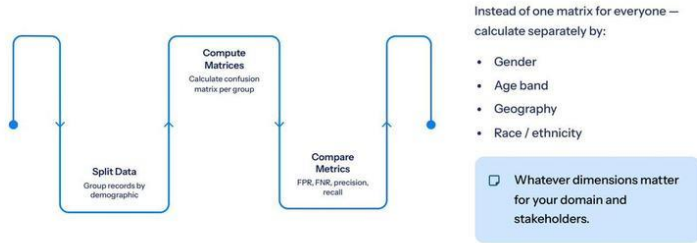
It becomes 92%.

Type the letter in the Zoom chat. Time to answer: 60 seconds.



V46 page 2: 85% Accuracy. Approved.

What Is Disaggregation?



V46 page 4: What Is Disaggregation?

Your turn: split the 85%

900 loan applicants in two groups. The model denies anyone it rates as high risk. Split its results by group and test them against a rule.

What you can change in the interactive

Show the table for

Options: All applicants; Group A only; Group B only

The same predictions, counted for everyone or for one group.

Group B's share of the test set

Options: 1 in 3: 600 A, 300 B (video); 1 in 5: 1,200 A, 300 B; 1 in 2: 600 A, 600 B

Changes how many records each group adds; each group's own results stay the same.

Largest gap the rule allows

Options: 5 points (the video's rule); 10 points; 15 points

The fairness rule is chosen before looking at the results.

Try, in this order

1. Switch the table to Group B only: accuracy 71%, and 21% of good payers wrongly denied.
2. Change Group B's share to 1 in 5: overall accuracy rises to 88%, yet nothing changed for Group B.
3. Raise the allowed gap to 15 points: the same model now passes. The rule changed, not the model.

The point: An overall number mixes groups by their size; split every result by group and fix the rule before you look.

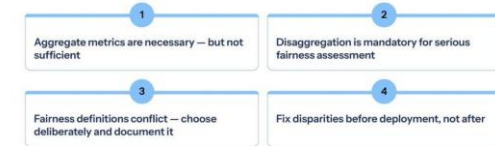
Made-up counts chosen to match the video's 85%, 92% and 71%. The video's separate example of 15% against 30% cannot sit with 92% accuracy, so this block uses 10% and 21%: still about twice as high for Group B.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

- 1 85% overall was 92% for 600 applicants and 71% for 300.
- 2 Good payers wrongly denied: 10% in A and 21% in B. Against a 5-point rule, the model fails.
- 3 Adding more Group A records pushed overall accuracy to 88% without changing anything for Group B.

Key Takeaways



V46 page 13: Key Takeaways

See the gap. Fix it first.

Disaggregated analysis turns hidden harm into an actionable engineering decision.

V46 page 14: See the Gap. Fix It First.

Up Next: Automate It

In the next video – specific tools and code to run disaggregated analysis at scale.



V46 page 15: Up Next: Automate It

Governance and accountability

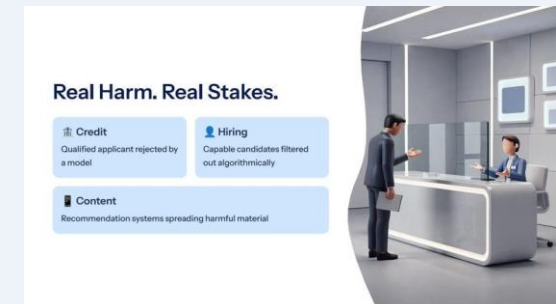
Video V47, Governance and Accountability in Machine Learning. 7 minutes.

When a model causes harm, someone must have approved it, watched it and been able to stop it.

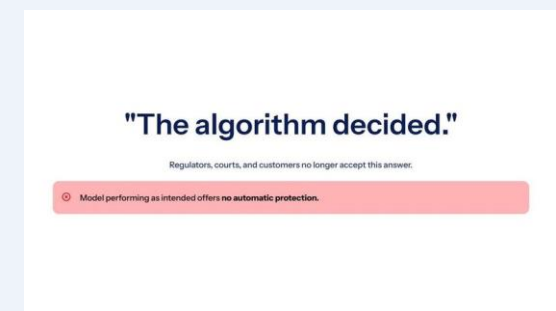
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 20 pages of V47.



V47 page 1: Governance and Accountability in Machine Learning



V47 page 2: Real Harm. Real Stakes.



V47 page 3: "The Algorithm Decided."

“The algorithm decided” is not an answer: courts ask who approved the model, who watched it and who could stop it.

No monitoring, no named owner

A bank's credit model passed a risk committee review, but in month 4 it starts treating one group worse: 20 extra wrongful denials a week. With no monitoring, a regulator complaint surfaces it after 26 weeks, and with no named owner, stopping takes 4 more: 600 applicants wrongly denied.

the word: **data drift**

Weekly checks and a named owner

The same drift is found by an automatic check within 2 weeks, and the named owner stops the model within a week: 60 applicants affected instead of 600.

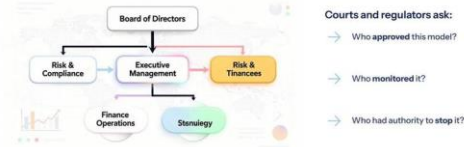
the word: **audit trail**

Clear ownership, documents written before launch and active monitoring: when any of the three is missing, legal and financial exposure follows.



V47 page 3: “The Algorithm Decided.”

What Accountability Actually Means



V47 page 4: What Accountability Actually Means

Realistic Scenario

Data Drift in Action

Bank deploys credit model Performs well at launch	Drift detected Worse outcomes for one demographic group emerge	Monitoring catches it in weeks Without it: a lawsuit or regulatory complaint surfaces the issue instead
--	---	--

V47 page 17: Realistic Scenario: Data Drift in Action

45 minutes with no way to stop, 2012

Step 1



New code for a new programme

On 1 August 2012, Knight Capital, a large US trading firm, prepared new trading software for a New York Stock Exchange programme for retail orders.

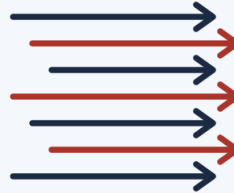
Step 2



One server of eight missed

Staff copied the update by hand to seven of its eight servers. The eighth still held old, unused code, and a reused setting switched that code on.

Step 3



Millions of unwanted orders

From the market opening, that server sent orders nobody intended. There was no automatic limit to stop it.

Step 4



45 minutes

Staff struggled to find the cause, and an early attempt to fix it made things worse. By the time the orders stopped, the firm had lost about \$440 million.

Step 5



Taken over and fined

The firm was taken over by a rival. In 2013 the SEC (Securities and Exchange Commission) fined it \$12 million for weak controls over its trading systems.

Sources: Knight Capital Group quarterly report (Form 10-Q) filed with the SEC for 2012; SEC order against Knight Capital Americas, Release No. 34-70694 (2013), as summarised in DEV Community and Medium case studies.

Three things to keep

1

Accountability means named people approved the model, monitor it and can stop it.

2

Privacy and anti-discrimination laws apply to models: an accurate model can still be illegal.

3

Three tools: a model card before launch, a risk committee review, and audit trails with monitoring after launch.

What Accountability Actually Means



V47 page 4: What Accountability Actually Means

Accurate ≠ Legal



V47 page 7: Accurate Is Not the Same as Legal

THREE CORE GOVERNANCE TOOLS

The Responsible ML Toolkit



V47 page 12: The Responsible ML Toolkit

A hiring model uses zip code and ends up screening out far more applicants of one race. Nobody put race into the data, and nobody intended the result.

Is the company legally exposed?

A No: there was no intent to discriminate.

B Yes: courts and regulators look at the outcome, not the intent.

C Only if the model's accuracy is low.

D No, because the model made the decision, not a person.

Case in Point

The Zip Code Problem

A hiring model trained on past employee data uses zip code — which correlates with race.

No explicit demographic input	Still violates equal-opportunity law	Removing the feature is not optional
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V47 page 8: Case in Point: The Zip Code Problem

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Your bank's credit model went live last month. A regulator asks who has the authority to stop it today, and nobody in the meeting can name that person.

What is missing?

- A** A larger training set.
- B** Clear ownership: a named person with the authority to pull the model.
- C** A more complex algorithm.
- D** A marketing plan for the model.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Governance Gap Check

If your organization deployed an ML model for credit decisions right now — could you name the person accountable if it caused harm?

If not, **that's a governance gap.**

V47 page 5: Governance Gap Check

Data drift adds 20 wrongful denials a week. A monthly review would find it after 5 weeks; a weekly automatic check after 2 weeks. The named owner stops the model 1 week after the problem is found.

How many applicants are wrongly denied with weekly checks, and with a monthly review?

A

40 and 100

B

60 and 120

C

20 and 20

D

120 and 60

Type the letter in the Zoom chat. Time to answer: 60 seconds.

Tool #3: Audit Trails & Monitoring

- Once live, track:
- 01 Is performance holding up over time?
 - 02 Is bias emerging in new data (data drift)?
 - 03 Are logs ready if something breaks?
 - 04 Can you pull the model quickly if needed?



V47 page 16: Tool 3: Audit Trails and Monitoring

Realistic Scenario

Data Drift in Action

Bank deploys credit model Performs well at launch	Drift detected Worse outcomes for one demographic group emerge	Monitoring catches it in weeks Without it: a lawsuit or regulatory complaint surfaces the issue instead
--	---	--

V47 page 17: Realistic Scenario: Data Drift in Action

Your turn: who notices, and who can stop it?

A bank's credit model. In month 4, data drift starts hurting one group. Choose the governance in place and count the applicants affected.

What you can change in the interactive

Before launch

Options: Nothing documented; A model card is written; Card and committee review

The committee review catches a zip code proxy before launch.

Monitoring after launch

Options: None; Monthly review; Weekly automatic check

Decides how soon a problem in live decisions is noticed.

A named person can stop the model

Tick box: no or yes

Without a named owner, deciding to stop takes 4 weeks instead of 1.

Try, in this order

1. Start with nothing in place: 900 applicants are wrongly denied before a regulator complaint surfaces the problem.
2. Add the risk committee review: the proxy is caught before launch and the count falls to 600.
3. Add weekly automatic checks and a named owner: 60 applicants, and all three non-negotiables are in place.

The point: Ownership, documents written before launch and active monitoring decide how many people a failing model reaches.

Fixed values for illustration: drift adds 20 wrongful denials a week, an unreviewed zip code proxy adds 10 more, and the weeks to detect and to stop are typical figures, not measurements.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

- 1 Courts look at outcomes, not intent: zip code standing in for race is still discrimination.
- 2 Without monitoring or an owner, drift reached 600 applicants; weekly checks and a named owner cut that to 60.
- 3 Write the governance evidence before launch; after harm occurs, it is too late to create it.



V47 page 9: Intent Doesn't Determine Legality



V47 page 18: The Three Non-Negotiables



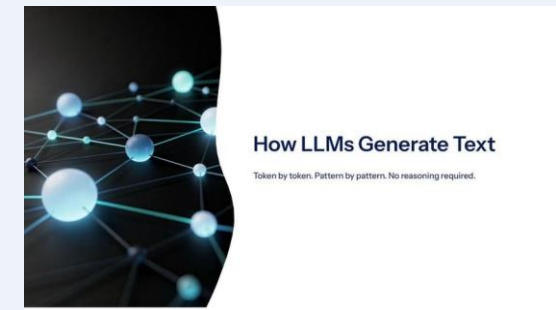
V47 page 19: Start Now

How LLMs write text

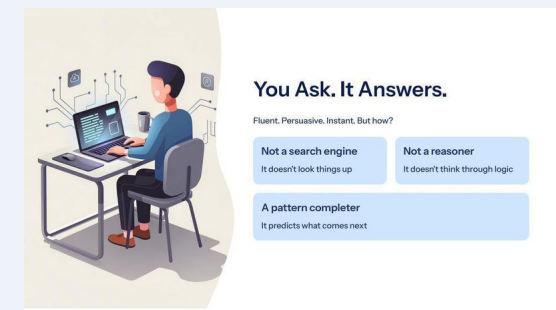
Video V48, How LLMs Generate Text. 10 minutes.

An LLM (large language model) writes by betting on the next token again and again; fluent text is not checked text.

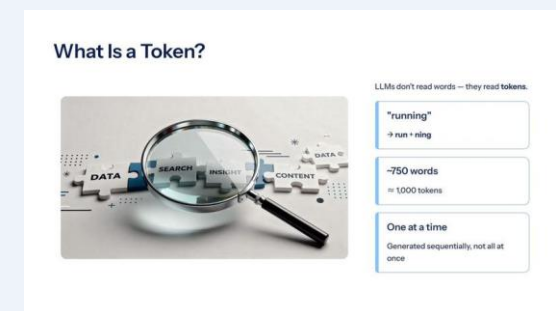
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 20 pages of V48.



V48 page 1: How LLMs Generate Text



V48 page 2: You Ask. It Answers.



V48 page 3: What Is a Token?

An LLM (large language model) writes by betting on the next token, one token at a time.

What it does not do

It does not look facts up or reason them through. For “The quick brown fox ____” it holds four bets learned from its training text: jumps 60%, runs 20%, walks 10%, other words 10%.

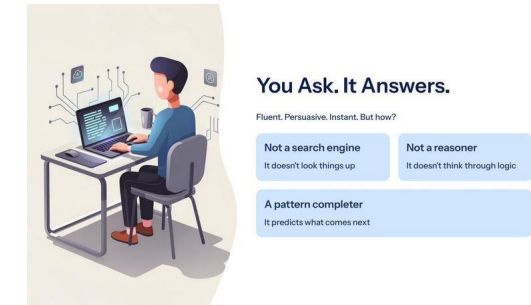
the word: **token**

What temperature changes

At temperature 0.5, “jumps” rises to 86%, so answers repeat. At 2.0 it falls to 42%, so answers vary. Neither setting makes the text true.

the word: **temperature**

Use it freely for drafts; verify anything that will be published or acted on.



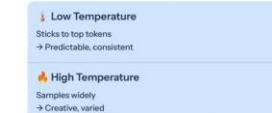
V48 page 2: You Ask. It Answers.

A Weighted Bet, Every Step



V48 page 5: A Weighted Bet, Every Step

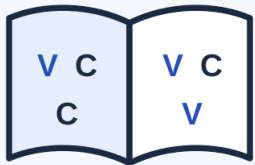
Introducing Temperature



V48 page 11: Introducing Temperature

Counting which letter comes next, 1913 and 1948

Step 1



20,000 letters by hand

In 1913 the Russian mathematician Andrey Markov took the first 20,000 letters of Pushkin's poem Eugene Onegin and marked each one as a vowel or a consonant.

Step 2



What follows what

After a vowel, the next letter was a vowel about 13 times in 100; after a consonant, about 66 times in 100. The previous letter changes the odds of the next one.

Step 3



Letters picked at random

In 1948 the American mathematician and engineer Claude Shannon showed that letters picked purely at random give text that looks like noise.

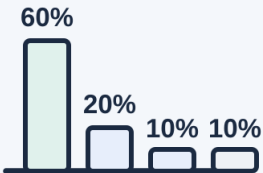
Step 4



More context, more English

When each new letter or word was picked using the odds of what came before, fragments that looked like English began to appear, without meaning anything.

Step 5



The same bet, far larger

Today's LLMs (large language models) make the same kind of bet, token by token, using much more context and billions of examples.

Sources: Purdue University course notes on Markov's 1913 study of Eugene Onegin; Communications of the ACM (Association for Computing Machinery), "Language Models: Past, Present, and Future"; C. E. Shannon, "A Mathematical Theory of Communication" (1948).

Three things to keep

- 1 LLMs read and write tokens, pieces of words, one at a time.
- 2 Each token is a bet: the likely next token, given everything written so far.
- 3 Temperature trades consistency for variety; neither setting checks facts.

What Is a Token?



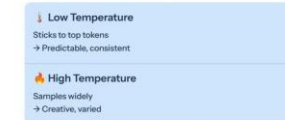
V48 page 3: What Is a Token?

One Job, Repeated



V48 page 4: One Job, Repeated

Introducing Temperature



V48 page 11: Introducing Temperature

A manager says the company's writing assistant “looked up our return policy and reasoned about it” before drafting a customer email. The assistant is a plain LLM with no connection to company files.

What actually happened?

- A** It searched the company's policy database.
- B** It predicted a likely sequence of tokens based on patterns in its training text.
- C** It checked the policy with a lawyer.
- D** It copied the policy word for word from the company website.

Type the letter in the Zoom chat. Time to answer: 45 seconds.



You Ask. It Answers.

Fluent. Persuasive. Instant. But how?

Not a search engine
It doesn't look things up

Not a reasoner
It doesn't think through logic

A pattern completer
It predicts what comes next

V48 page 2: You Ask. It Answers.

A retailer needs 500 product descriptions written in the same style, with no surprises from one product to the next.

Which settings fit best?

- A** High temperature, 2.0, for variety.
- C** Temperature does not matter for product descriptions.

- B** Low temperature or greedy picking, for consistent and repeatable wording.
- D** A different random temperature for each product.

Temperature in Practice



Low Temperature

Product descriptions, summaries, structured reports — reliability wins



High Temperature

Tagline brainstorm, campaign ideas, creative drafts — novelty wins

V48 page 12: Temperature in Practice

Type the letter in the Zoom chat. Time to answer: 45 seconds.

At temperature 2.0 with sampling, 10 runs of “The quick brown fox ____” give jumps 4 times, runs 2 times, walks 2 times and other words 2 times.

What do 10 runs give with greedy picking?

A Jumps 6 times, runs 2, walks 1, other words 1.

B Jumps all 10 times.

C A different word in every run.

D The same mix as with sampling: jumps 4, runs 2, walks 2, other words 2.

A Weighted Bet, Every Step



"The quick brown fox ____" — the model picks the most likely next word.



V48 page 5: A Weighted Bet, Every Step

The Loop in Action



Each chosen token becomes part of the next prediction's context. The output builds itself, one token at a time.

V48 page 6: The Loop in Action

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Your turn: the next word, one bet at a time

The model has read “The quick brown fox” and holds four bets for the next word. Change the temperature and the way the word is picked.

What you can change in the interactive

Temperature

Options: 0.5 (low); 1.0 (standard); 2.0 (high)

Low makes the likely word even more likely; high spreads the chances out.

How the next word is picked

Options: Most likely word (greedy); By chance (sampling)

Greedy always gives the same answer; sampling can differ from run to run.

Run number

Slider from 1 to 10

Each run uses its own fixed draw from 1 to 100, so the results repeat exactly.

Try, in this order

1. Keep greedy picking and change the temperature: the chances move, but every run still gives “jumps”.
2. Switch to sampling at temperature 1.0 and move through the runs: “jumps” wins 5 of 10, and run 2 picks “walks”.
3. Try temperature 0.5, then 2.0: “jumps” wins 8 of 10 at low temperature and only 4 of 10 at high temperature.

The point: Every word is a weighted bet; temperature trades consistency for variety, and neither checks whether the text is true.

The 60%, 20%, 10% and 10% chances come from the video; “other words” stands for everything else combined. The ten draws are fixed numbers so the runs repeat exactly; a real tool draws new numbers every time.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

1

The fox example: jumps 60%, runs 20%, walks 10%, other words 10%.

2

Low temperature pushed “jumps” to 86%; high temperature cut it to 42%.

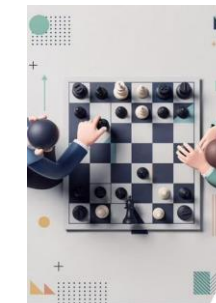
3

Fluent is not factual: review anything legal, medical, financial or public before it goes out.

The Power vs. The Risk

⚡ The Power	⚠ The Risk
• Fluent, contextual output • Feels natural and human • Accelerates drafting massively	• Confident-sounding hallucinations • No self-verification • False facts delivered with conviction

V48 page 13: The Power vs. The Risk



Fluent ≠ Expert

LLMs can sound expert on topics they've never verified — because the **pattern** fit, not because the **fact** is true.
Understanding this separates smart users from everyone else.

V48 page 17: Fluent Is Not Expert

Key Takeaways

 Token-by-token prediction Every output is a sequence of statistical bets	 Pattern, not knowledge Fluent ≠ factual. No reasoning, no retrieval
 Temperature = your creativity dial Low for consistency. High for variety.	 Human review is non-negotiable For anything consequential — always verify

V48 page 19: Key Takeaways

Hallucination and checking

Video V49, Hallucination and LLMs. 8 minutes.

An invented fact reads exactly like a real one, so the depth of checking must follow the stakes.

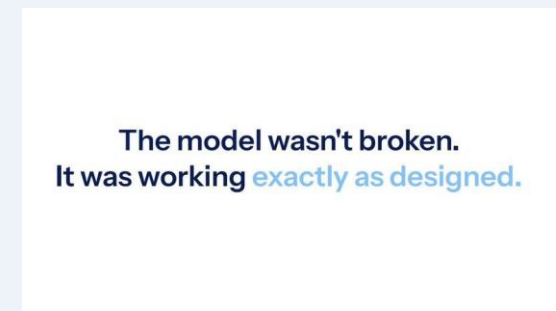
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 20 pages of V49.



V49 page 1: Hallucination and LLMs



V49 page 2: A Story About Fake Case Law



V49 page 3: The Model Wasn't Broken

An invented fact reads exactly like a checked one, so check according to what is at stake.

Headline numbers only

A 30-claim market report hides 3 invented claims: 1 headline number and 2 supporting details. Checking the 10 headline numbers takes 40 minutes and catches 1. That suits a leadership deck, not a regulator filing.

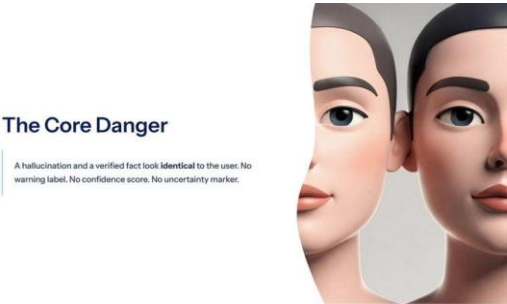
the word: **hallucination**

Every factual claim

Checking all 30 claims takes 120 minutes and catches all 3. Tier 3 outputs, such as a compliance filing, need that level, because one wrong penalty threshold can mean a fine.

the word: **verification**

A confident tone tells you nothing; where the output goes decides how much you check.



V49 page 5: The Core Danger

The Three-Tier Verification Framework



V49 page 11: The Three-Tier Verification Framework

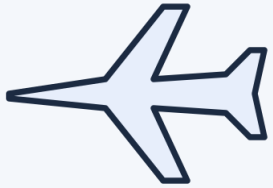
Tier 2 in Action



V49 page 13: Tier 2 in Action

Six court decisions that did not exist, 2023

Step 1



A routine injury case

A passenger sued the airline Avianca in a New York federal court, saying a serving cart hurt his knee on a 2019 flight.

Step 2



Research by chatbot

To answer the airline's request to dismiss the case, one of his lawyers asked a chatbot for earlier decisions. It supplied six, with names, citations and quotations.

Step 3



Nobody could find them

The airline's lawyers searched legal databases and found none of the six. The judge asked for copies.

Step 4



More invented text

Instead of checking, the lawyers filed extracts that the chatbot had also invented.

Step 5



Sanctions, June 2023

On 22 June 2023, Judge P. Kevin Castel fined the two lawyers and their firm \$5,000 and ordered them to write to the real judges wrongly named as authors.

Sources: Mata v. Avianca, Inc., US District Court for the Southern District of New York, sanctions opinion of 22 June 2023, as summarised by LegalClarity and the Legal AI (artificial intelligence) Governance case tracker.

Three things to keep

- 1 Hallucinations are plausible, fluent and false, and they carry no warning label.
- 2 LLMs are built for fluent text, not accuracy, and their training data contains errors and outdated facts.
- 3 Three tiers: brainstorm gets normal review, internal strategy gets spot checks, and legal, financial, medical or public outputs get every claim checked.

What Is Hallucination?



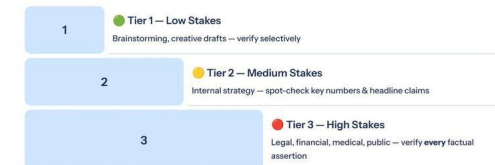
V49 page 4: What Is Hallucination?

Optimized for **Plausibility** — Not Accuracy



V49 page 8: Optimised for Plausibility, Not Accuracy

The Three-Tier Verification Framework



V49 page 11: The Three-Tier Verification Framework

A manager asks an LLM for legal support on a contract clause. It returns three court precedents with full citations, case numbers and confident detail. Two of the three cases never existed.

What went wrong?

- A** The model was broken.
- B** The model worked as designed: it produced text that fits the pattern of legal citations, with no step that checks facts.
- C** The manager asked the question the wrong way.
- D** The two cases were deleted from court records.

Type the letter in the Zoom chat. Time to answer: 45 seconds.



A Story About Fake Case Law

- Manager needs legal support for a contract clause
- LLM returns 3 precedents — full citations, case numbers, confident detail
- Two of those cases **never existed**

V49 page 2: A Story About Fake Case Law

This week an LLM helps with four outputs: a brainstorm of campaign slogans, an internal draft email to the team, a regulatory compliance filing, and personal notes from a meeting.

Which output needs every factual claim checked against an official source?

- A The brainstorm of campaign slogans.
- C The regulatory compliance filing citing rules, penalties and deadlines.

- B The internal draft email to the team.
- D The personal notes from the meeting.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

The Three-Tier Verification Framework



V49 page 11: The Three-Tier Verification Framework

A report going to a regulator has 30 factual claims and hides 3 invented ones: 1 headline number and 2 supporting details. The team checks only the 10 headline numbers, at 4 minutes each.

What is the result?

A 40 minutes, 1 invented claim caught and 2 left in: not enough for a tier 3 output.

B 40 minutes, and all 3 invented claims caught.

C 120 minutes, and 1 invented claim caught.

D No time spent and nothing caught, which is fine for a regulator.

Type the letter in the Zoom chat. Time to answer: 60 seconds.

Tier 2 in Action



Market research for leadership

- LLM suggests growth rates, competitor metrics
- Pull original research reports
- Verify headline numbers before presenting

Spot-check. Don't skip it.

V49 page 13: Tier 2 in Action

Tier 3 in Action



Regulatory compliance filing

- Every cited regulation → verified against official source
- Every penalty threshold → confirmed
- Every deadline → cross-checked

One error = audit or fine. Full verification is non-negotiable.

V49 page 14: Tier 3 in Action

Your turn: check by what is at stake

An LLM drafted a market report. Some claims are invented and read exactly like the real ones. Choose where it goes and how much you check.

What you can change in the interactive

Where the output goes

Options: Tier 1: internal brainstorm; Tier 2: leadership deck; Tier 3: regulator filing

The destination sets the stakes, and the stakes set the checking.

What you check

Options: Normal reading only; The headline numbers; Every factual claim

Each claim takes 4 minutes to check against an original source.

Report length

Options: 30 factual claims; 60 factual claims

A third of the claims are headline numbers; one claim in ten is invented.

Try, in this order

1. Tier 2 with normal reading only: 0 of 3 invented claims caught, which is not enough for a leadership deck.
2. Check the headline numbers: 40 minutes and 1 invented claim caught, enough for tier 2.
3. Change the destination to tier 3: the same 40 minutes are not enough; checking every claim takes 120 minutes and catches all 3.

The point: Invented claims look like real ones, so the destination, not the tone, decides how much you check.

A made-up report for illustration: 1 invented headline number and 2 invented details in every 30 claims, 4 minutes per check. Normal reading catches none, because invented claims read like true ones.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

1

Two of the three citations in the manager's answer did not exist, and they looked real.

2

Checking 10 headline numbers caught 1 of 3 invented claims in 40 minutes; checking all 30 caught all 3 in 120.

3

If an output will be acted on, shared outside, or used without expert review, verify it.

Your Verification Decision Rule



V49 page 16: Your Verification Decision Rule

The Core Insight



V49 page 18: The Core Insight

Key Takeaways

- Hallucinations are structural — not accidental
- Confidence tone ≠ factual accuracy
- Match your verification depth to your stakes
- Build the system *before* you need it



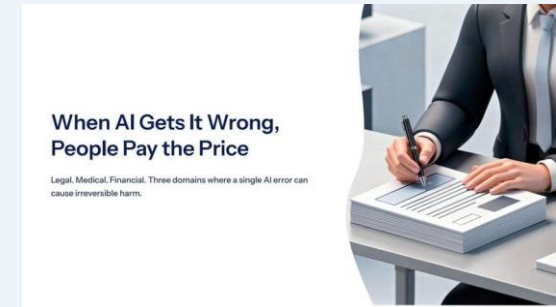
V49 page 20: Key Takeaways

When AI gets it wrong

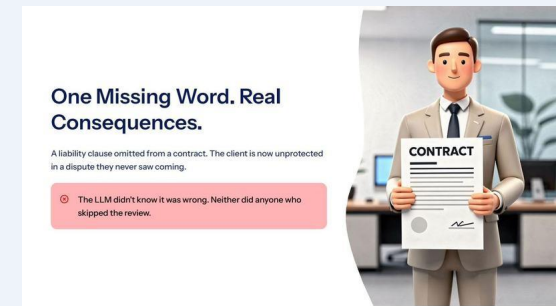
Video V50, When AI Gets It Wrong, People Pay the Price. 8 minutes.

In legal, medical and financial work a person must review AI (artificial intelligence) output, and that review has to be planned, paid for and recorded.

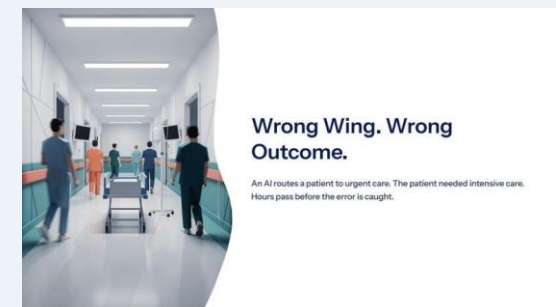
In this block: the one idea, a story, three things to keep, three questions, your turn (the interactive, described here), and three things from the block. The speaker notes carry a plain-words note for all 28 pages of V50.



V50 page 1: When AI Gets It Wrong, People Pay the Price



V50 page 2: One Missing Word. Real Consequences.



V50 page 3: Wrong Wing. Wrong Outcome.

Human review costs money; skipping it costs more, and the liability stays with a person, never the model.

No review of 200 contract drafts

About 2% contain a serious error, such as a missing liability clause: 4 drafts a month. At about \$50,000 per dispute, that is \$200,000 of exposure.

the word: **liability**

A lawyer checks every draft

45 minutes each at \$300 an hour: 150 hours and \$45,000 a month, and the 4 errors are caught before a client signs. The review is the smaller number in the budget.

the word: **human-in-the-loop**

Plan the review before launch: who checks, what they check, what authority they have, how much time is budgeted, and how each decision is recorded.

One Missing Word. Real Consequences.

A liability clause omitted from a contract. The client is now unprotected in a dispute they never saw coming.

The LLM didn't know it was wrong. Neither did anyone who skipped the review.



V50 page 2: One Missing Word. Real Consequences.

Three Domains. One Shared Rule.

Legal
Lawyers are personally liable for contract failures

Medical
Clinicians and hospitals are accountable for patient outcomes

Financial
Advisors are responsible for every recommendation made

The cost of error exceeds the cost of human review – every single time.

V50 page 8: Three Domains. One Shared Rule.

Review Has a Cost. Budget It.



Review takes time. Account for it.

- Legal doc → lawyer hours budgeted
- Medical triage → clinician sign-off staffed
- Financial output → analyst time scheduled

Build it into your project plan and budget from day one.

V50 page 17: Review Has a Cost. Budget It.

The airline and its chatbot, 2022 to 2024

Step 1



A question after a death

In November 2022, after his grandmother died, Jake Moffatt asked the chatbot on Air Canada's website about bereavement fares.

Step 2



A confident wrong answer

The chatbot said he could buy full-price tickets and claim the lower fare within 90 days. The airline's own policy page, which the chatbot linked to, said the opposite.

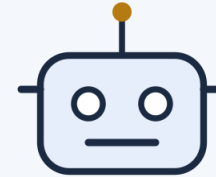
Step 3



Claim refused

He paid about CA\$1,630 for the return trip to Toronto and applied in time. Air Canada refused, while admitting the chatbot had used misleading words.

Step 4



"The chatbot is responsible"

At British Columbia's Civil Resolution Tribunal, the airline argued that the chatbot was a separate legal entity responsible for its own actions.

Step 5



The company answers for it

On 14 February 2024 the tribunal rejected that: the chatbot is part of Air Canada's website, and the company is responsible for what it says. Air Canada had to pay CA\$812.02.

Sources: Moffatt v. Air Canada, 2024 BCCRT 149 (British Columbia Civil Resolution Tribunal); CBC News, February 2024.

Three things to keep

- 1

Legal, medical and financial errors can be irreversible, and liability cannot be handed to a model.
- 2

A reviewer is named, knows what to verify, and can change, reject or escalate the output.
- 3

Budget the review time and record every decision, or auditors will assume no review happened.

Irreversible.

These aren't operational glitches. A patient's outcome changed. A client's legal standing weakened. An investor's capital is at risk.

Legal Client unprotected	Medical Patient misrouted	Financial Capital exposed
------------------------------------	-------------------------------------	-------------------------------------

V50 page 5: Irreversible.

Liability Cannot Be Delegated

A lawyer puts their name on a contract. If it fails, they can be sued. The law does not allow liability transfer to a language model.



V50 page 7: Liability Cannot Be Delegated

Defining the Review Step

1	2	3
Who Reviews Lawyer, clinician, or senior analyst — named and accountable	What They Verify Citations, reasoning validity, regulatory compliance	Their Authority Can they modify, reject, or escalate the output?

V50 page 16: Defining the Review Step

A law firm uses an LLM (large language model) to draft contracts. One draft leaves out a liability clause, a partner signs it off without noticing, and the client loses a dispute.

Who is accountable?

A The company that sells the LLM.

B The model itself.

C The lawyer who signed off, and the firm.

D Nobody, because the draft was automated.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

One Missing Word. Real Consequences.

A liability clause omitted from a contract. The client is now unprotected in a dispute they never saw coming.

⊗ The LLM didn't know it was wrong. Neither did anyone who skipped the review.



V50 page 2: One Missing Word. Real Consequences.

Liability Cannot Be Delegated

A lawyer puts their name on a contract. If it fails, they can be sued. The law does not allow liability transfer to a language model.



V50 page 7: Liability Cannot Be Delegated

A financial team produces 200 advice summaries a month with an AI tool. About 3% contain a serious error costing about \$10,000 each. An analyst needs 15 minutes per summary, at \$120 an hour.

What does reviewing every summary cost per month, and what does it prevent?

A \$6,000 of review prevents about \$60,000 of errors.

B \$60,000 of review prevents about \$6,000 of errors.

C \$24,000 of review prevents about \$60,000 of errors.

D The two cannot be compared.

Type the letter in the Zoom chat. Time to answer: 60 seconds.

FINANCIAL DOMAIN

LLM Failure Modes in Finance

Outdated Data

Training cutoffs miss market shifts

Hallucinated Sources

Cited with full confidence

Repealed Regulations

Cited as if still in force

A human advisor verifies:

- Current market data
- Recent regulatory guidance
- Verifiable, existing sources

V50 page 12: Financial Domain: LLM Failure Modes

Review Has a Cost. Budget It.



Review takes time. Account for it.

- Legal doc → lawyer hours budgeted
- Medical triage → clinician sign-off staffed
- Financial output → analyst time scheduled

Build it into your project plan and budget from day one.

V50 page 17: Review Has a Cost. Budget It.

A hospital will start using an AI triage tool next month. The project plan covers accuracy testing but says nothing about human review.

What should be decided before launch?

A

Nothing yet; handle problems when they appear.

B

Who reviews (a named clinician), what they check, their authority to change or reject, the time budgeted, and how each decision is recorded.

C

Only the tool's accuracy target.

D

Whether patients like the tool.

Type the letter in the Zoom chat. Time to answer: 45 seconds.

Map Review Points Before You Launch

Ask three questions about every AI-assisted workflow:

1. Where does accuracy matter most?
2. Where would an error cause serious harm?
3. Who reviews, and what do they check?



V50 page 15: Map Review Points Before You Launch

Your turn: budget the review

An AI tool drafts outputs in a high-stakes domain. Choose how much a qualified person reviews, and compare the review bill with the errors that get through.

What you can change in the interactive

Domain

Options: Legal: contract drafts; Medical: triage suggestions; Financial: advice summaries

Each domain has its own reviewer, review time and cost of an error.

Human review

Options: No review; Half of the outputs checked; Every output checked

A reviewer catches the serious error in every output they check.

AI outputs per month

Options: 200; 400; 1,000

Review hours and errors both grow with volume.

Try, in this order

1. Legal with no review: 4 serious errors reach clients, \$200,000 of exposure.
2. Every output checked: 150 lawyer hours and \$45,000, and no error gets through.
3. Medical at 1,000 outputs: 50 clinician hours, or 10 serious errors reach patients.

The point: Review has a price you can budget; the errors it prevents cost more, and the accountability stays with a named person.

Illustrative values: 45 minutes at \$300 an hour for a lawyer, 15 minutes at \$120 for an analyst, 3 minutes at \$150 for a clinician; serious errors in 2%, 3% and 1% of outputs; \$50,000 per legal error and \$10,000 per financial error. Harm to patients is counted, not priced.

In the HTML edition you move these controls, and the numbers, the sentence and the “What just changed” panel update after every change.

Three things from this block

1

200 contract drafts with no review: 4 serious errors and \$200,000 of exposure; full review: \$45,000.

2

Four governance questions: who is accountable, what the reviewer must see, which errors the model will miss, and whether review is documented.

3

Design the review step before deployment, not after the first error.

The Four Governance Questions



V50 page 20: The Four Governance Questions

Design for Review from Day One



V50 page 25: Design for Review from Day One



The Right Approach, Summarized

1. Identify review points **before** deployment
2. Name the reviewer, define their authority
3. Budget the time and staffing
4. Document every decision with a clear audit trail

V50 page 27: The Right Approach, Summarised

Before Thursday 24 September

- Take Quizzes 4 and 5 (Modules 7 to 10), 15 minutes each, before Session 5.
- Sit the midterm exam (Modules 1 to 6, 90 minutes online) in the 18 to 23 September window.
- Watch videos V51 to V57 before Session 5.
- Submit the AI Solution Evaluation Matrix (Module 11) before Session 5; the same template works for your capstone.
- Start choosing your capstone business problem; it is needed before Session 6 on 26 September.

Next session: Session 5, Thursday 24 September 2026, 11:00 AM to 12:30 PM EDT: designing for deployment, and deciding whether an AI (artificial intelligence) project is worth it. Videos V51 to V57.

What is assessed, and when

What	Weight	When
Quizzes 4 and 5 (Modules 7 to 10)	Part of 10%	Before 24 September 2026
Midterm exam (Modules 1 to 6, 90 minutes online)	30%	Between 18 and 23 September 2026
AI Solution Evaluation Matrix (Module 11), counts for participation	Part of 10%	Before 24 September 2026
Quiz 6 (Modules 11 and 12)	Part of 10%	Before 26 September 2026
Capstone project brief	50%	Draft 1 Oct, peer reviews 4 Oct, revision 8 Oct, final 10 Oct 2026



V50 page 27: The Right Approach, Summarised



V50 page 28: What Comes Next